

Computing

- Computing technology is ubiquitous, in the form of computers, tablets, smartphones, the web, cloud computing, email, text messages, social media, and much more.



Cloud Computing

- Cloud computing is an Information Technology (IT) paradigm that enables ubiquitous access to **shared pools of configurable resources** system and **higher-level services** that can be rapidly provisioned minimal management effort, often over the Internet with Cloud computing relies on sharing of resources to achieve coherence and economies of scale, similar to a public utility.

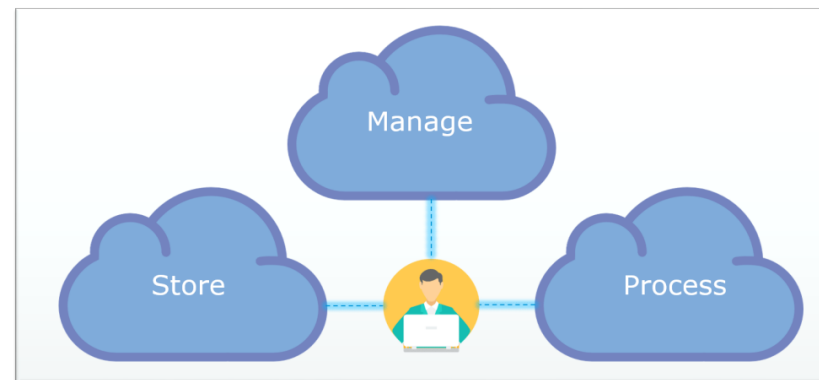
-Wikipedia

Cloud Computing

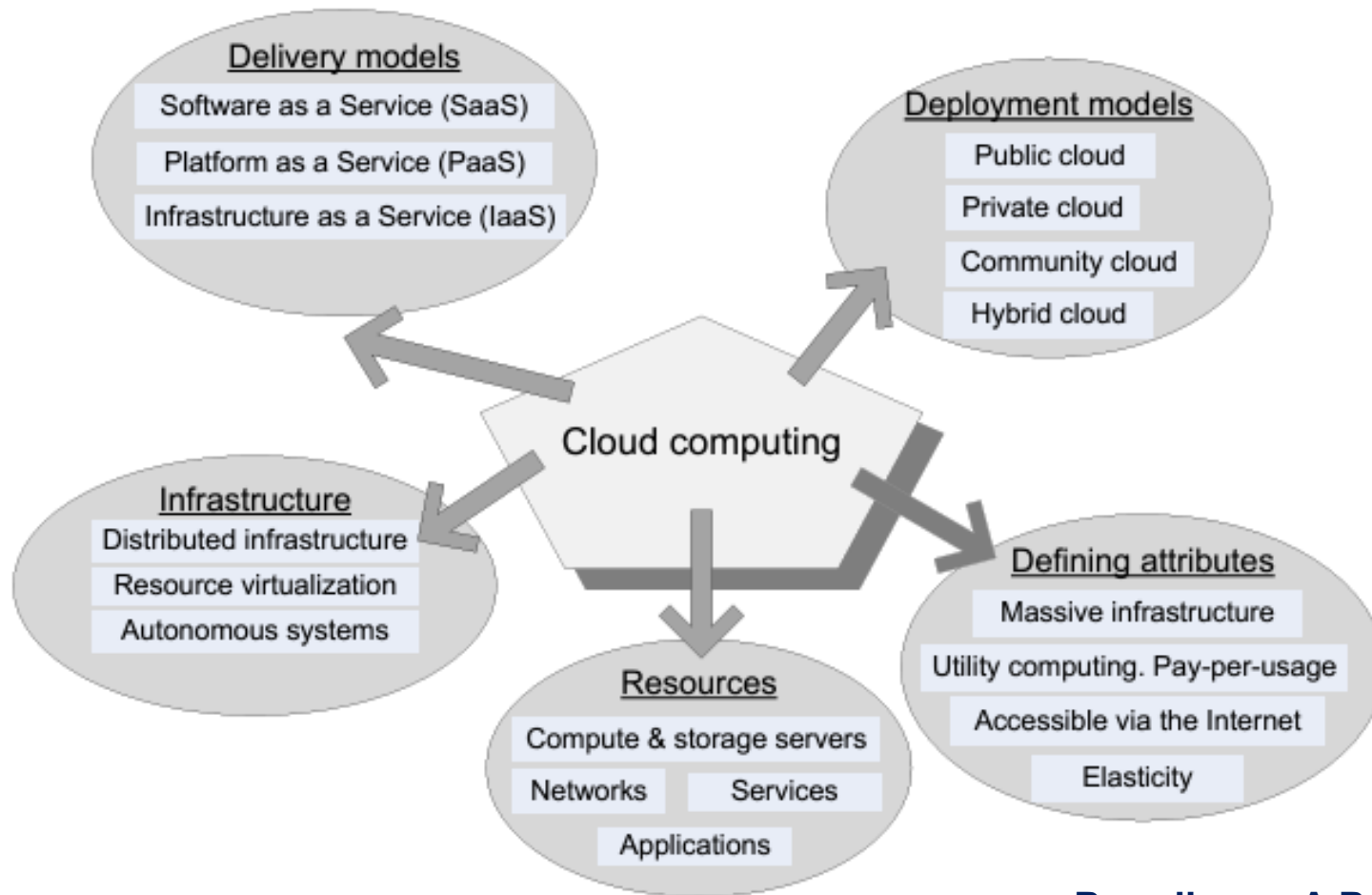
- Simply put, cloud computing is the delivery of computing services servers, storage, databases, networking, software, analytics and more over the Internet (“the cloud”).
- Companies offering these computing services are called cloud providers and typically charge for cloud computing services based on usage, similar to how user billed for gas or electricity at home.

Cloud Computing

- It is the use of remote servers on the internet to store, manage and process data rather than a local server or user personal computer.
 - **Store:** Store large or small files on the cloud, which user can access on the go!
 - **Manage:** Manage user data with optimized databases on the cloud.
 - **Process:** With scalable compute power on the cloud, user can process any amount of data in moments!
- It is a type of internet computing which allows users to access the services on pay per use basis.



Cloud Computing – Paradigm



Paradigm – A Pattern or Model

Early Models of Cloud Computing

- Basic reasoning: information and data processing can be done more efficiently on large farms of computing and storage systems accessible via the Internet.
- Two early models:
 1. **Grid computing** – initiated by the National Labs in the early 1990s; targeted primarily at scientific computing.
 - *"Grid computing is the collection of computer resources from multiple locations to reach a common goal. The grid can be thought of as a distributed system with non-interactive workloads that involve a large number of files."* from Wikipedia
 2. **Utility computing** – initiated in 2005-2006 by IT companies and targeted at enterprise computing.
 - *"Utility computing is a service provisioning model in which a service provider makes computing resources and infrastructure management available to the customer as needed, and charges them for specific usage rather than a flat rate."* from Wikipedia

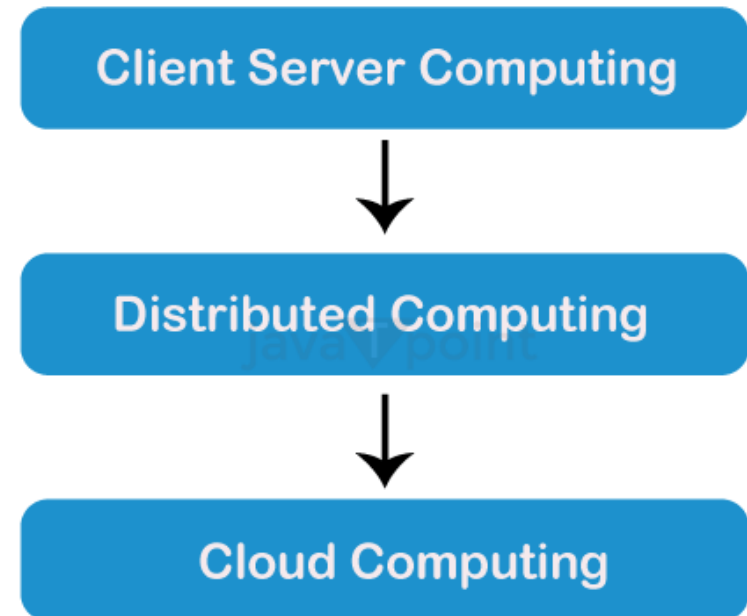
History of Cloud Computing

- Cloud computing is all about renting computing services.
- This idea first came in the 1950s.
- In making cloud computing what it is today, five technologies played a vital role. These are distributed systems and its peripherals, virtualization, web 2.0, service orientation, and utility computing.



History of Cloud Computing

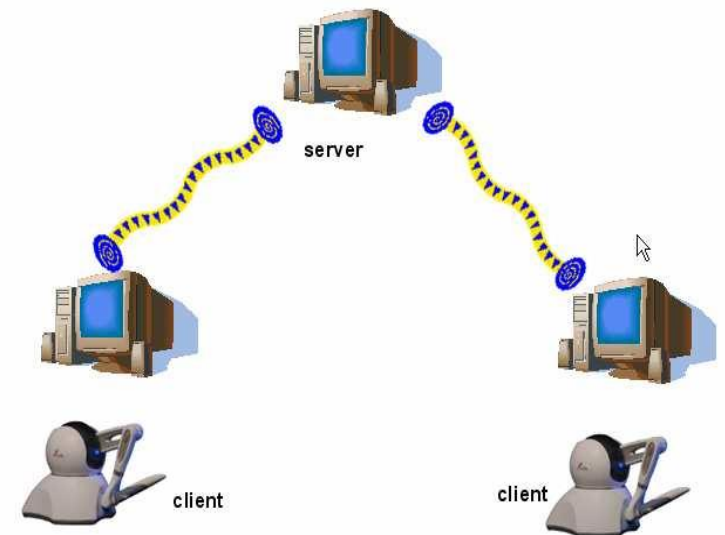
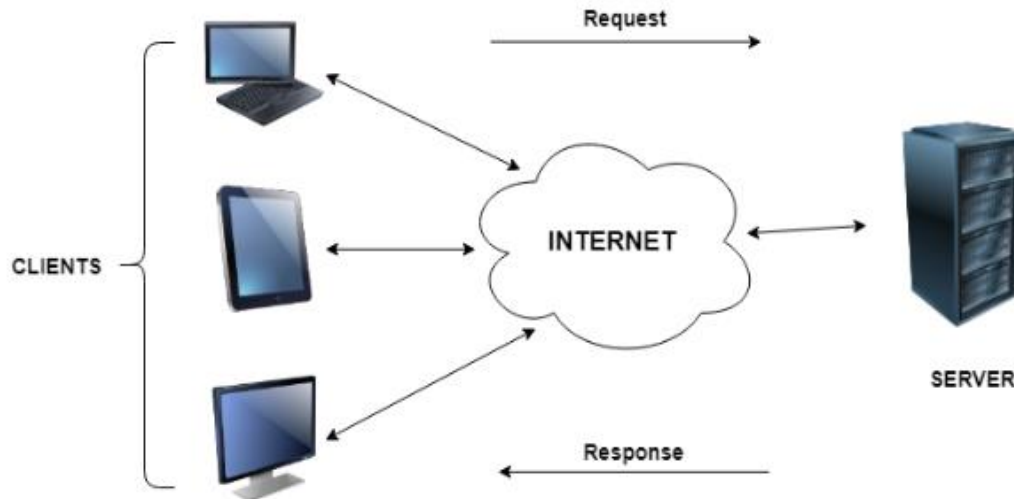
- Since the beginning days of computing, when mainframe computers were accessible remotely through terminals, "cloud computing" has evolved.
- However, with the development of internet technologies and the demand for more effective and scalable computing solutions in the 1990s and early 2000s, the contemporary idea of cloud computing as we know it today first emerged.



History of Cloud Computing

Client – Server Computing

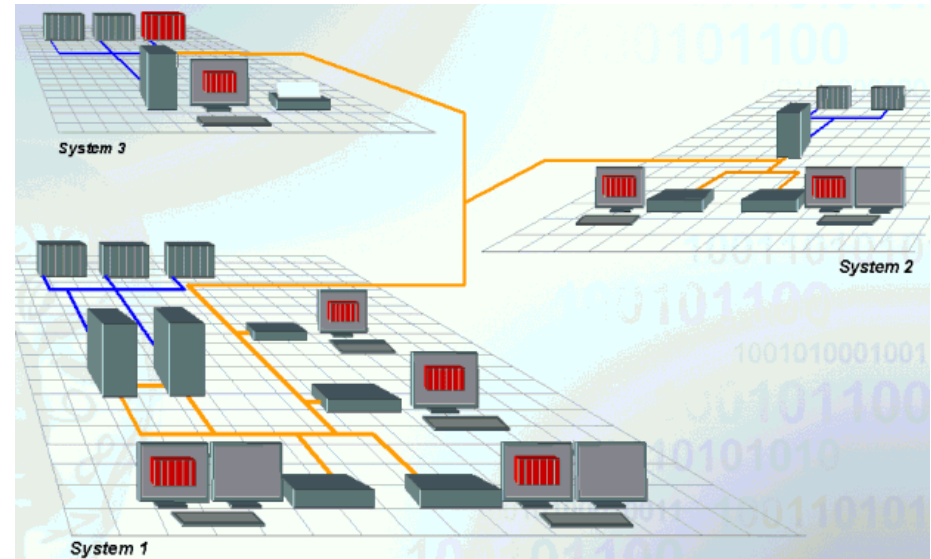
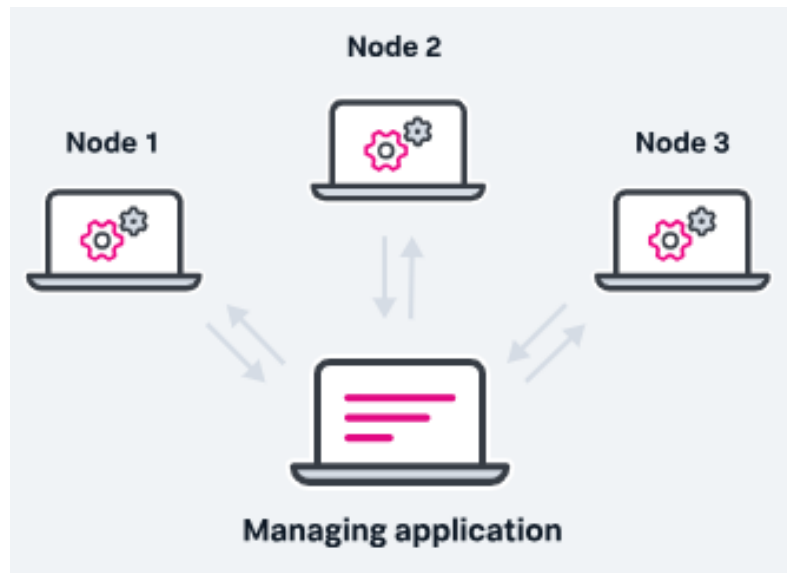
- In client server computing, the clients requests a resource and the server provides that resource.
- A server may serve multiple clients at the same time while a client is in contact with only one server.
- Both the client and server usually communicate via a computer network but sometimes they may reside in the same system



History of Cloud Computing

Distributed Computing

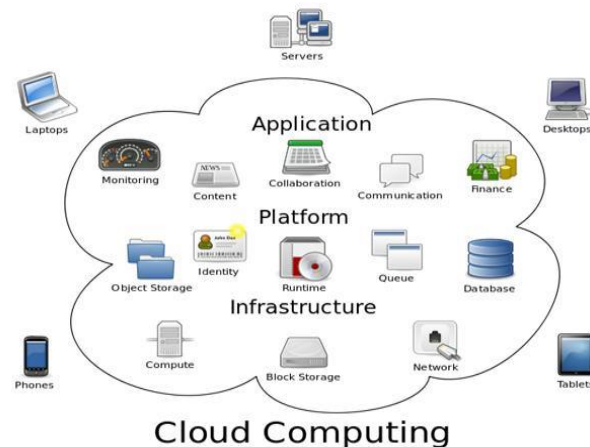
- Distributed computing is the method of making multiple computers work together to solve a common problem.
- It makes a computer network appear as a powerful single computer that provides large-scale resources to deal with complex challenges



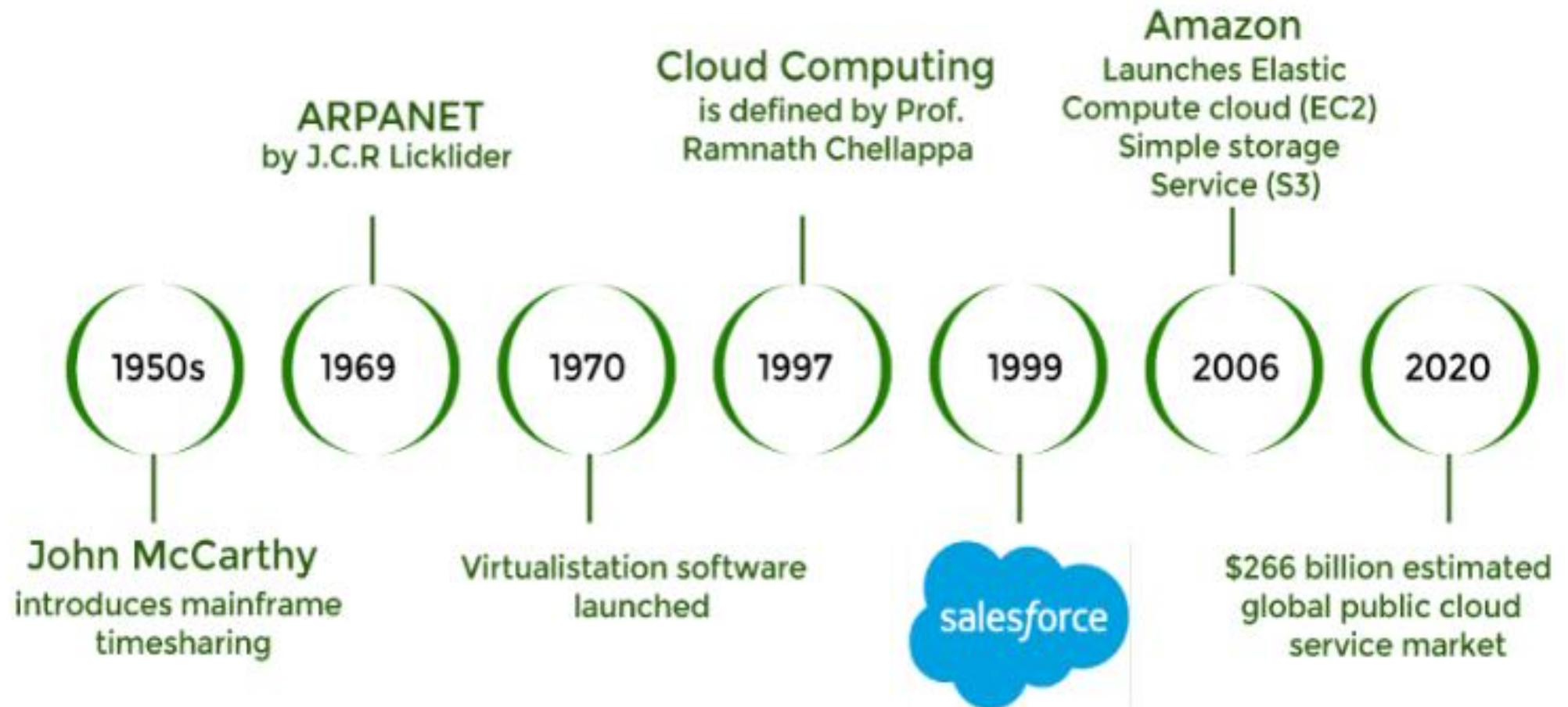
History of Cloud Computing

Cloud Computing

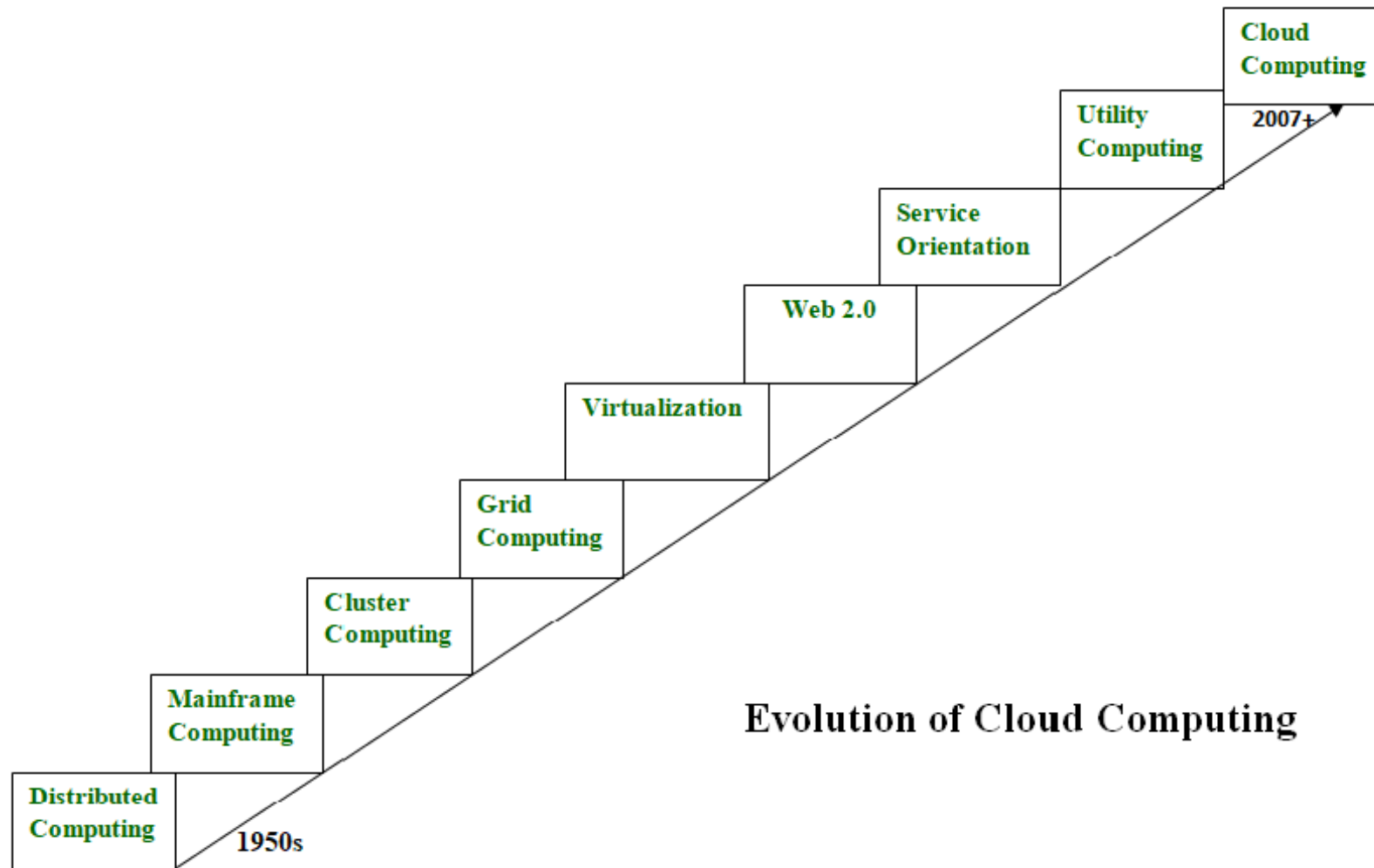
- Client/server and distributed computing paradigms served as the cornerstones for the paradigm that eventually developed as cloud computing.
- The objective was to offer network-based, primarily the Internet, on-demand access to shared computer resources and services.
- The goal of cloud computing was to offer consumers flexible, scalable, and economical access to computing resources, storage, and applications. It shifted the emphasis to distant services and pay-as-user-go business models from local infrastructure and ownership.



Cloud Computing History



Evolution of Cloud Computing



Evolution of Cloud Computing

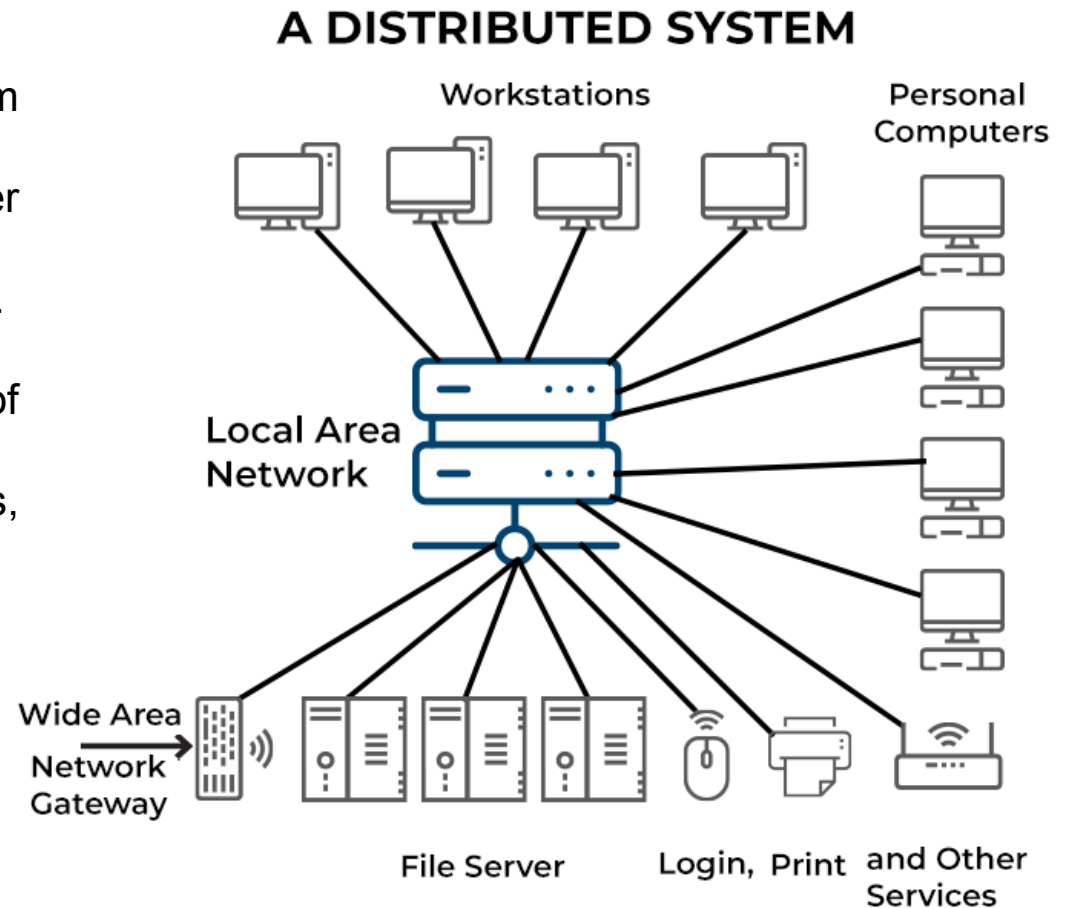
Evolution of Cloud Computing

Distributed Systems

- It is a composition of multiple independent systems but all of them are depicted as a single entity to the users. The purpose of distributed systems is to share resources and also use them effectively and efficiently.
- Distributed systems possess characteristics such as scalability, concurrency, continuous availability, heterogeneity, and independence in failures. But the main problem with this system was that all the systems were required to be present at the same geographical location

Evolution of Cloud Computing

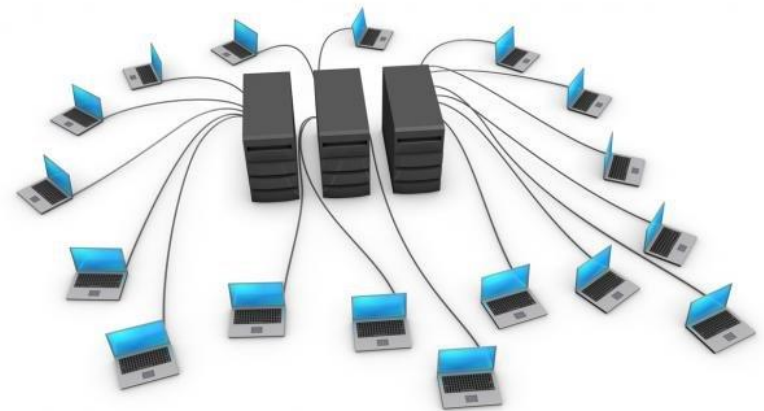
- Distributed computing is defined as a system consisting of software components spread over different computers but running as a single entity.
- A distributed system can be an arrangement of different configurations, such as mainframes, computers, workstations, and minicomputers.



Evolution of Cloud Computing

Mainframe Computing

- Mainframes which first came into existence in 1951 are highly powerful and reliable computing machines. These are responsible for handling large data such as massive input-output operations. Even today these are used for bulk processing tasks such as online transactions etc.
- These systems have almost no downtime with high fault tolerance. After distributed computing, these increased the processing capabilities of the system. But these were very expensive. To reduce this cost, cluster computing came as an alternative to mainframe technology.



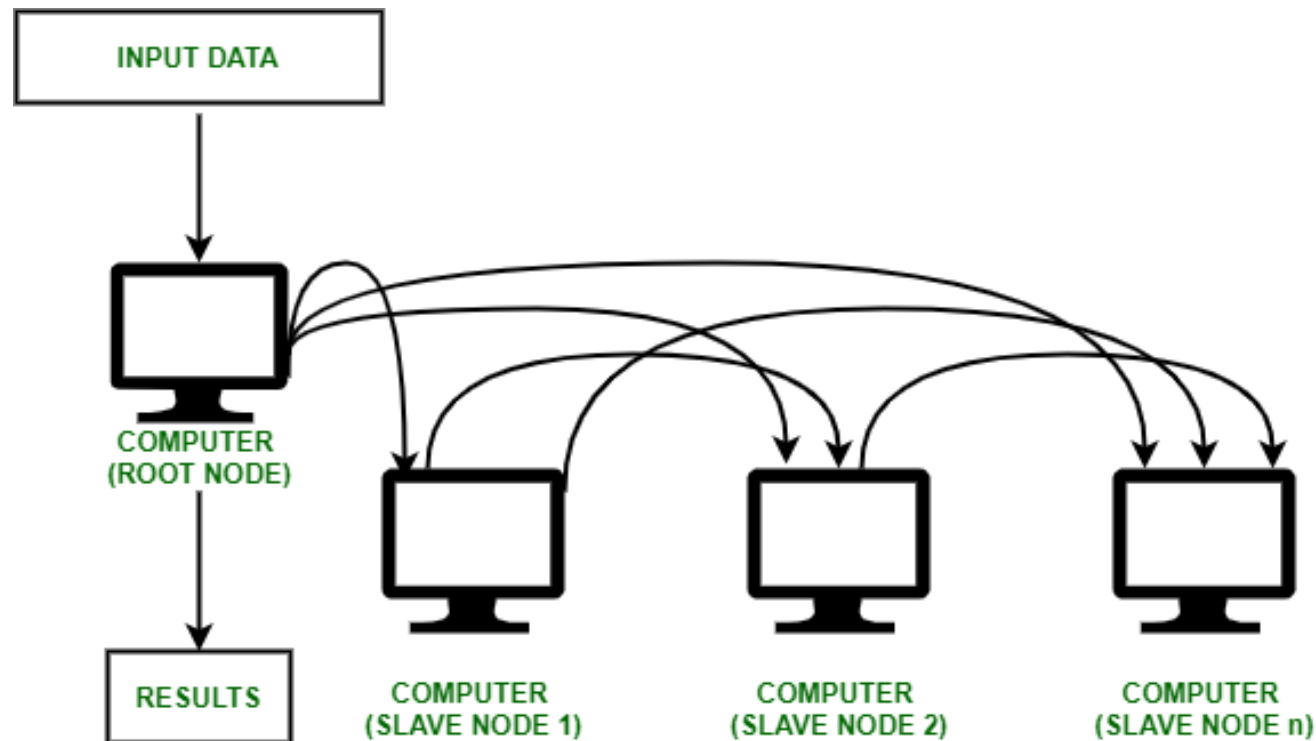
Evolution of Cloud Computing

Cluster Computing

- In 1980s, cluster computing came as an alternative to mainframe computing.
- Each machine in the cluster was connected to each other by a network with high bandwidth.
- These were way cheaper than those mainframe systems.
- These were equally capable of high computations. Also, new nodes could easily be added to the cluster if it was required.
- Thus, the problem of the cost was solved to some extent but the problem related to geographical restrictions still pertained. To solve this, the concept of grid computing was introduced

Evolution of Cloud Computing

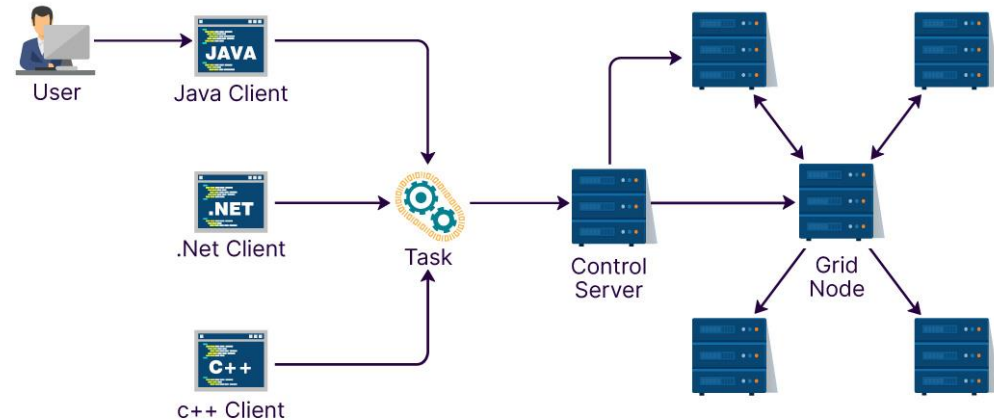
Cluster Computing



Evolution of Cloud Computing

Grid Computing

- In 1990s, the concept of grid computing was introduced. It means that different systems were placed at **entirely different geographical locations and these all were connected via the internet.**
- These systems belonged to different organizations and thus the grid consisted of heterogeneous nodes. Although it solved some problems but new problems emerged as the **distance between the nodes increased.**

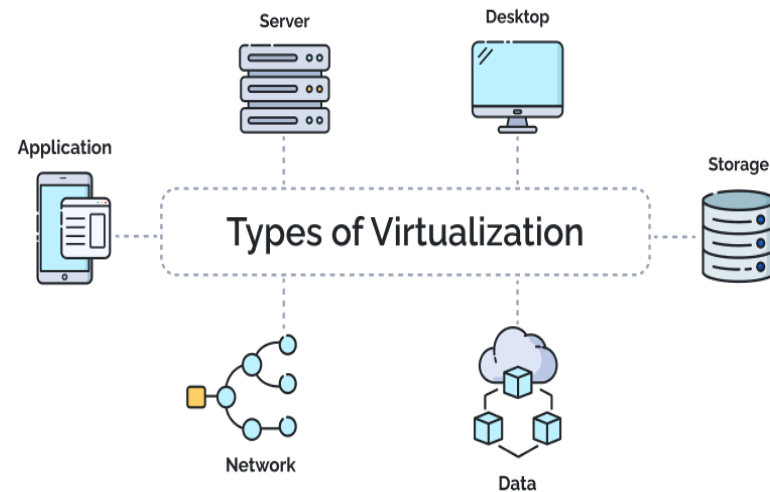


HOW GRID COMPUTING WORKS

Evolution of Cloud Computing

Virtualization

- It was introduced nearly 40 years back. It refers to the process of creating a virtual layer over the hardware which allows the user to run multiple instances simultaneously on the hardware. It is a key technology used in cloud computing. It is the base on which major cloud computing services such as Amazon EC2, VMware vCloud, etc work on. Hardware virtualization is still one of the most common types of virtualization.

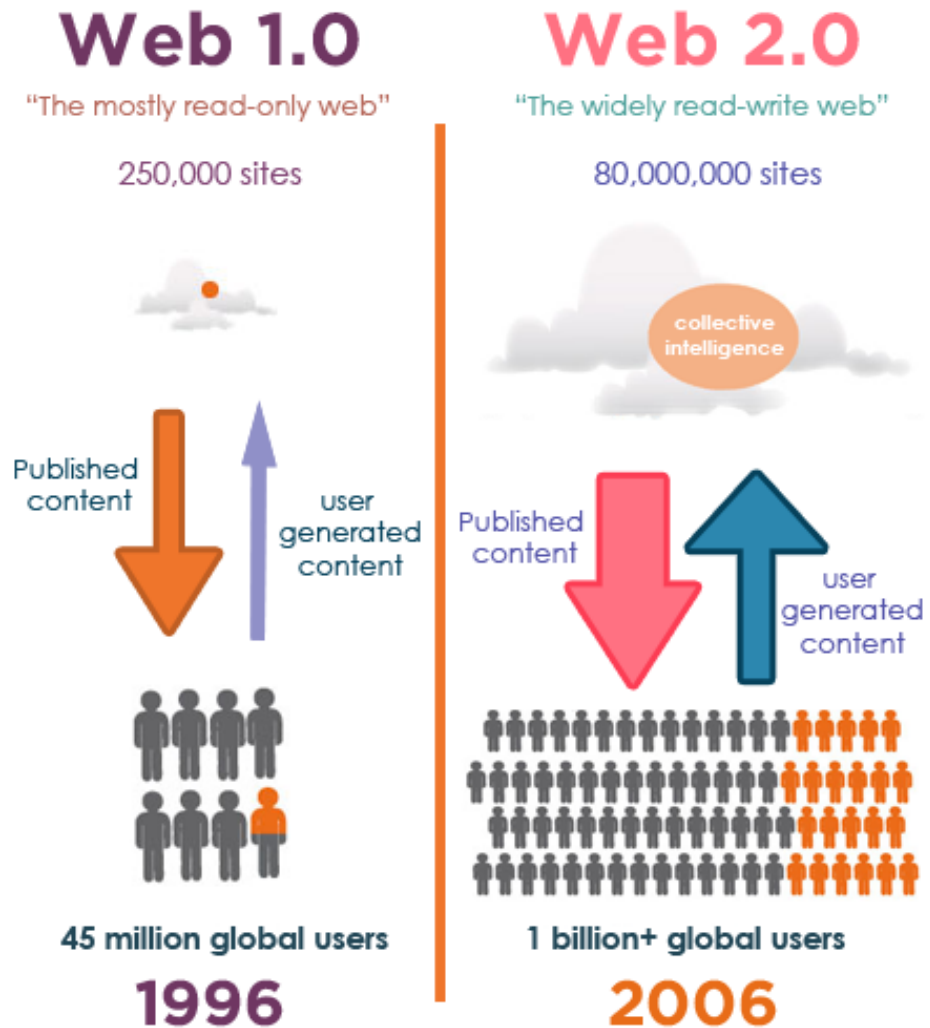


Evolution of Cloud Computing

Web 2.0

- It is the interface through which the cloud computing services interact with the clients.
- It is because of Web 2.0 that we have interactive and dynamic web pages.
- It also increases flexibility among web pages. Popular examples of web 2.0 include Google Maps, Facebook, Twitter, etc.
- Needless to say, social media is possible because of this technology only.
- It gained major popularity in 2004.

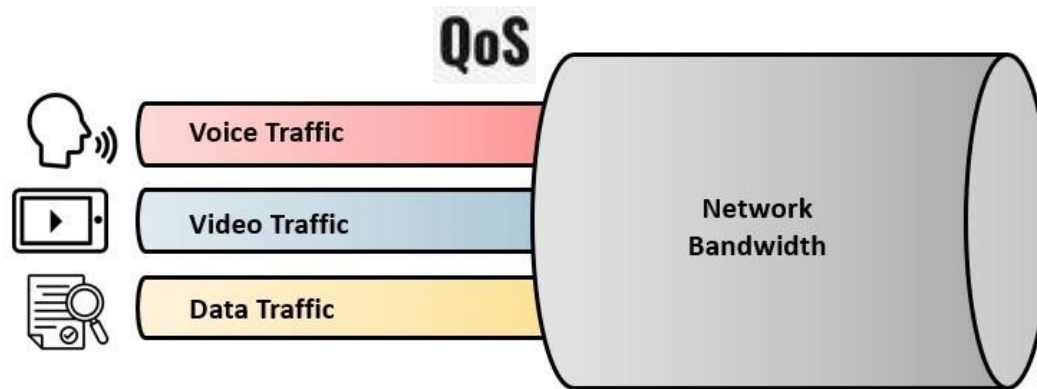
Evolution of Cloud Computing



Evolution of Cloud Computing

Service Orientation

- It acts as a reference model for cloud computing. It supports low-cost, flexible, and evolvable applications. Two important concepts were introduced in this computing model. These were Quality of Service (QoS) which also includes the SLA (Service Level Agreement) and Software as a Service (SaaS)



Evolution of Cloud Computing

Utility Computing

- Utility computing, also known as pay-per-use or metered services, is a service provisioning model that gives customers access to computing resources and infrastructure management on an on-demand basis.

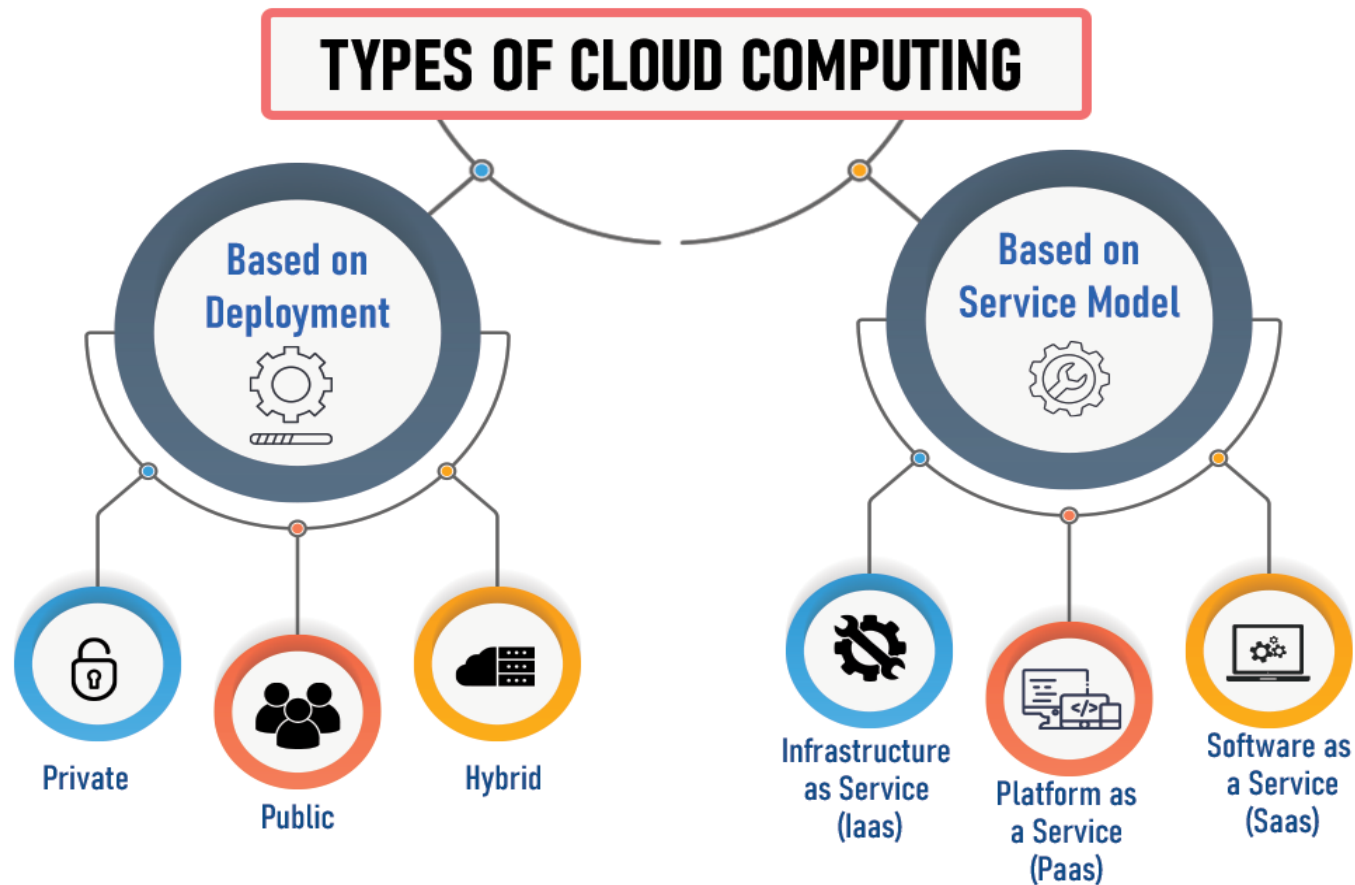


On Premises vs. Cloud Computing

	On-premises	Cloud
SECURITY	IT teams are responsible for the security of all data and workloads.	Providers manage technical administration of security and offer tools for vulnerability management.
RELIABILITY	Even the strongest data center will break at some point. Systems failures and natural disasters can take down entire facilities.	Outages affect availability, so apps that require high availability should stay on premises or be extended across regions for additional redundancy.
COST	Companies pay for the hardware—plus excess capacity if needed—as well as facility operations.	Only pay for what is used. But companies can overpay if a workload isn't properly optimized.
EXPERTISE	Those who have studied and worked with these technologies know the ins and outs of the infrastructure, tape library management, required programming languages and so on.	Takes away the more complex and challenging parts of computing, which allows IT teams to home in on business objectives. It also requires learning skills to adapt to the new environment.



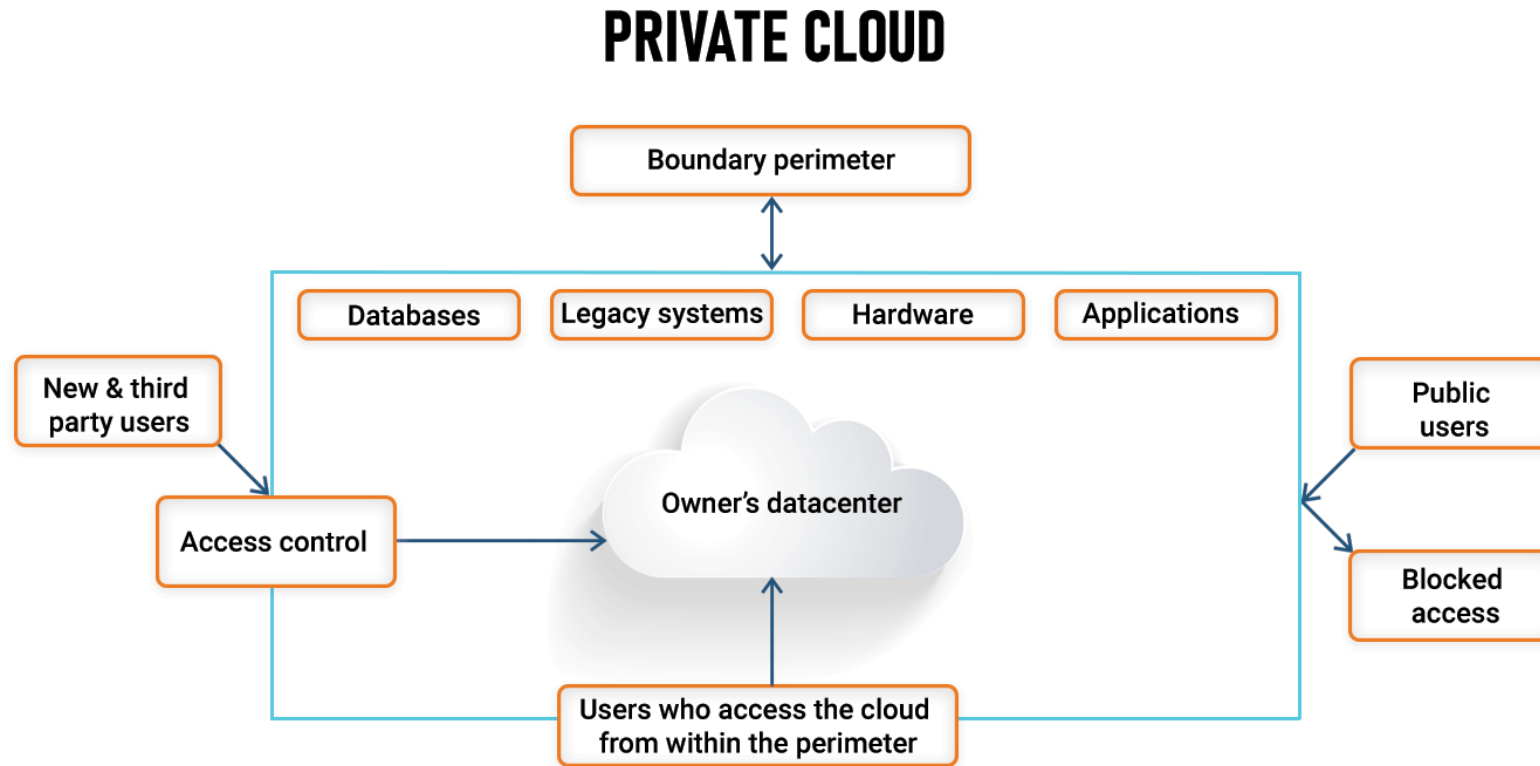
Types of Cloud Computing



Private Cloud

- In a private cloud, the computing services are offered over a private IT network for the dedicated use of a single organization.
- Also termed internal, enterprise, or corporate cloud, a private cloud is usually managed via internal resources and is not accessible to anyone outside the organization.
- Private cloud computing provides all the benefits of a public cloud, such as self-service, scalability, and elasticity, along with additional control, security, and customization.

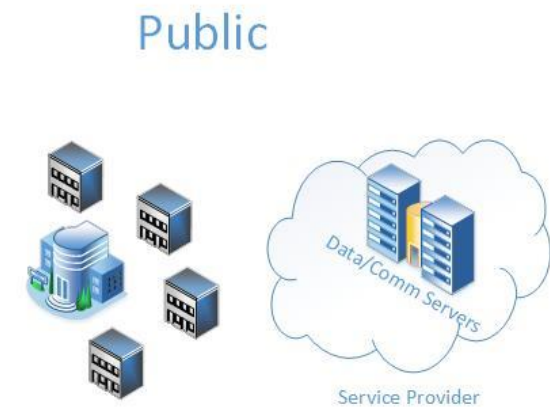
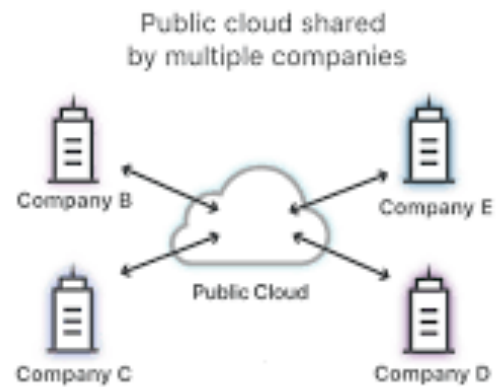
Private Cloud Computing



Public Cloud Computing

- Public cloud refers to computing services offered by third-party providers over the internet. Unlike private cloud, the services on public cloud are available to anyone who wants to use or purchase them.
- These services could be free or sold on-demand, where users only have to pay per usage for the CPU cycles, storage, or bandwidth they consume.
- Public clouds can help businesses save on purchasing, managing, and maintaining etc.

Public Cloud Computing



Hybrid Cloud Computing

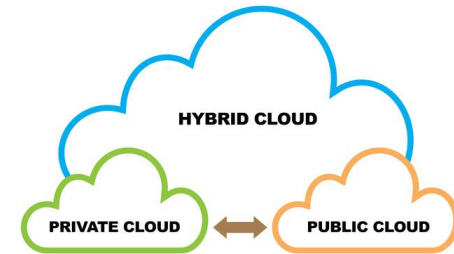
- Hybrid cloud uses a combination of public and private cloud features. The “best of both worlds” cloud model allows a shift of workloads between private and public clouds as the computing and cost requirements change.
- When the demand for computing and processing fluctuates, hybrid cloud opens a new window allows businesses to scale their on-premises infrastructure up to the public cloud to handle the overflow while ensuring that no third-party data centers have access to their data.

Hybrid Cloud Computing



HYBRID CLOUD

- Combination of both public and private cloud
- Shared security responsibility
- Helps maintain tighter controls over sensitive data and processes



PUBLIC CLOUD

- Offered by third-party providers
- Available to anyone over the public internet
- Scales quickly and convenient



PRIVATE CLOUD

- Offered to select users over the internet or a private internal network
- Provides greater security controls
- Requires traditional datacenter staffing and maintenance

Comparison

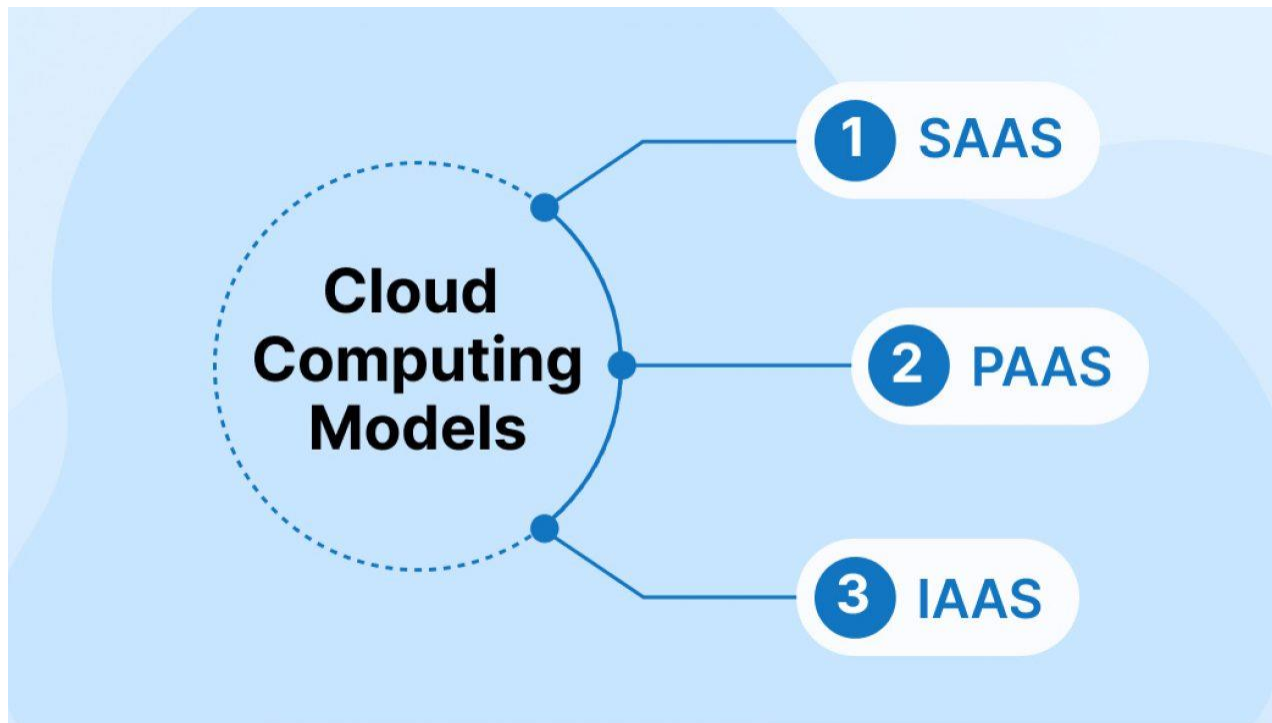
Parameters\Type	Public Cloud	Private Cloud	Hybrid Cloud	Community Cloud
Description	In public cloud, services are available for public users.	Private cloud is build up with existing private infrastructure. This type of cloud has some authentic users who can dynamically provision the resources.	Hybrid cloud is a heterogeneous distributed system, resulting from a private cloud, which incorporates different types of services and resources from public clouds.	Different types of cloud are integrated together to meet a common or particular need for some organizations.
Scalability	Very High	Limited	Very High	Limited
Reliability	Moderate	Very High	Medium to High	Very High
Security	Totally Depends on service provider	High class security	Secure	Secure
Performance	Low to medium	Good	Good	Very Good
Cost	Cheaper	High Cost	Costly	Costly
Examples	Amazon EC2, Google AppEngine	VMWare, Microsoft, KVM, Xen	IBM, HP, VMWare vCloud, Eucalyptus	SolaS Community Cloud, VMWare

Comparison – Advantages and Disadvantages

		
Public Cloud	Hybrid Cloud	Private Cloud
- Services are owned and operated by a third party provider. - The maintenance is bared by the service provider. - Pay-as-you-go model. Thus, the setting and operating cost is less. - Lesser security as the platform is shared. - Lesser flexibility & control over the cloud environment.	- Often called as 'the best of both worlds', it combines both public & private cloud. - Greater flexibility & more deployment options. - Cloud bursting is also possible. - Network complexities & compliance issues. - Can be extremely expensive.	- Dedicated to a single organization. - Higher security as the resources are not shared. - Greater flexibility to control the cloud environment. - Purchase and maintenance has to be bared by the organization - Expensive than public cloud.

Cloud Delivery Models

1. **Software as a Service (SaaS)** (high level)
2. **Platform as a Service (PaaS)**
3. **Infrastructure as a Service (IaaS)** (low level)

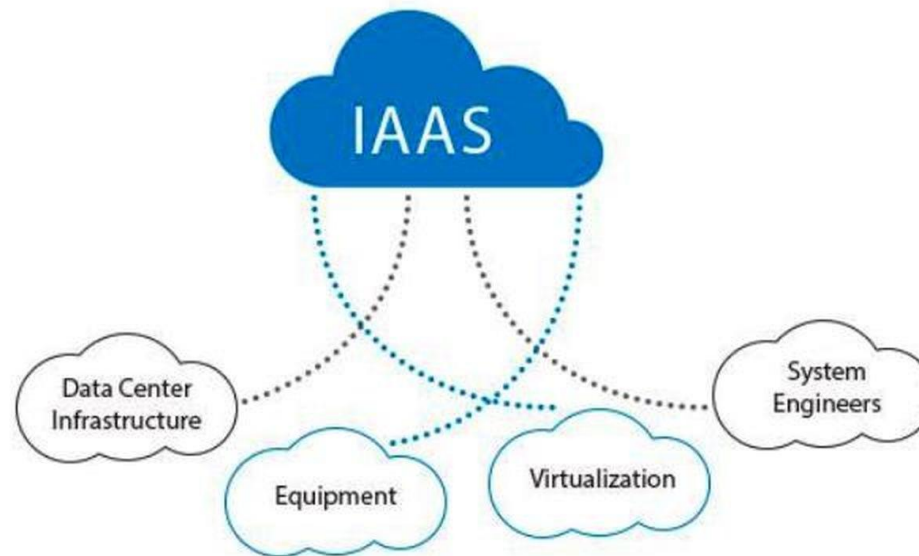


Infrastructure as a Service (IaaS)

- Infrastructure is compute resources, CPU, VMs, storage, etc
- The user is able to deploy and run arbitrary software, which can include operating systems and applications.
- The user does not manage or control the underlying Cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of some networking components, e.g., host firewalls.
- Services offered by this delivery model include: server hosting, storage, computing hardware, operating systems, virtual instances, load balancing, Internet access, and bandwidth provisioning.

Infrastructure as a Service (IaaS)

- Infrastructure as a service or IaaS is a type of cloud computing in which a service provider is responsible for **providing servers, storage, and networking over a virtual interface**.



Infrastructure as a Service (IaaS)

Characteristics:

- Resources are available as a service
- Services are highly scalable
- Dynamic and flexible
- GUI and API-based access
- Automated administrative tasks



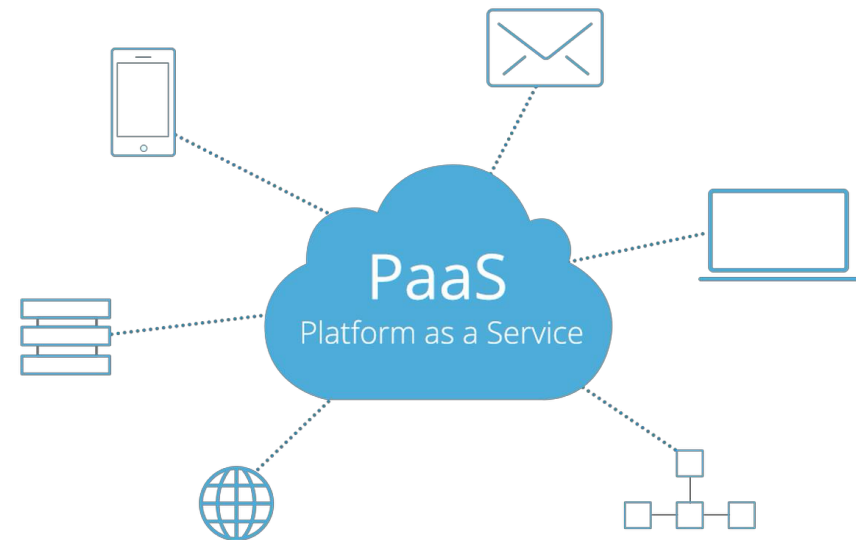
Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.

Platform as a Service (PaaS)

- Allows a cloud user to deploy consumer-created or acquired applications using programming languages and tools supported by the service provider.
- The user:
 - Has control over the deployed applications and, possibly, application hosting environment configurations.
 - Does not manage or control the underlying Cloud infrastructure including network, servers, operating systems, or storage.
- Not particularly useful when:
 - The application must be portable.
 - Proprietary programming languages are used.
 - The hardware and software must be customised to improve the performance of the application.
- Examples: Google App Engine, Windows Azure

Platform as a Service (PaaS)

- Platform as a service or PaaS is a type of cloud computing that provides a development and deployment environment in cloud that allows users to develop and run applications without the complexity of building or maintaining the infrastructure.
- It provides users with resources to develop cloud-based applications.
- In this type of service, a user purchases the resources from a vendor on a pay-as-user-go basis and can access them over a secure connection.



Platform as a Service (PaaS)

Characteristics

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- Support multiple languages and frameworks.
- Provides an ability to "Auto-scale".

Example: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift

Software as a Service (SaaS)

- Applications are supplied by the service provider.
- The user does not manage or control the underlying Cloud infrastructure or individual application capabilities.
- Services offered include:
 - Enterprise services such as: workflow management, communications, digital signature, customer relationship management (CRM), desktop software, financial management, geo-spatial, and search.
- Not suitable for real-time applications or for those where data is not allowed to be hosted externally.
- Examples: Gmail, Salesforce

Software as a Service (SaaS)

- SaaS or software as a service allows users to access a vendor's software on cloud on a subscription basis. In this type of cloud computing, users don't need to install or download applications on their local devices.
- Instead, the applications are located on a remote cloud network that can be directly accessed through the web or an API.

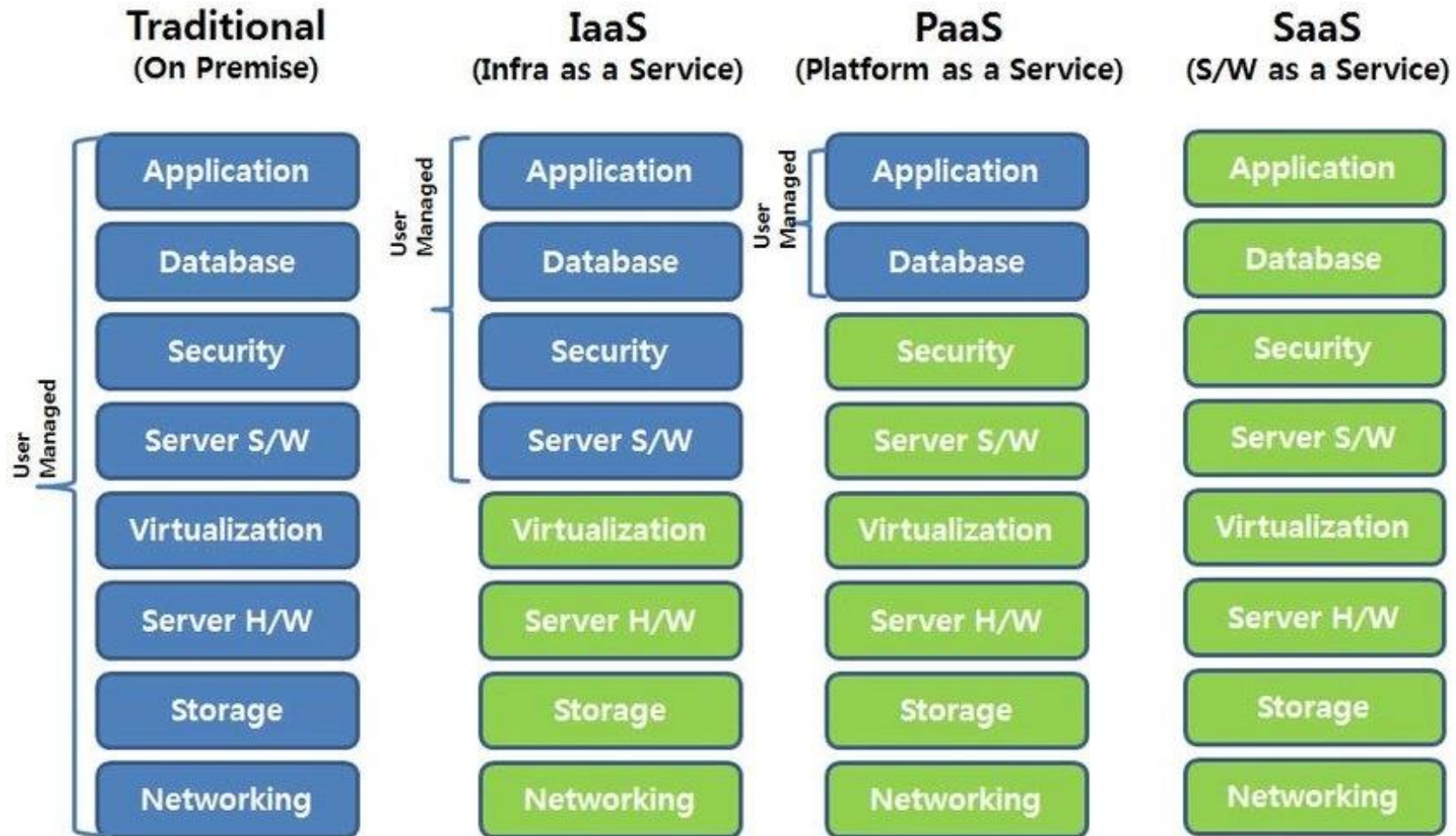


Software as a Service (SaaS)

Characteristics

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis
- **Example:** BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

Comparison



IaaS, PaaS and SaaS

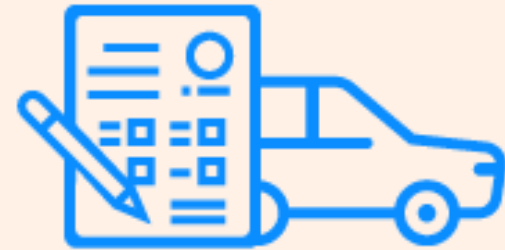
YOUR OWN CAR

On-premises solution



LEASED CAR

IaaS



TAXI

PaaS



BUS

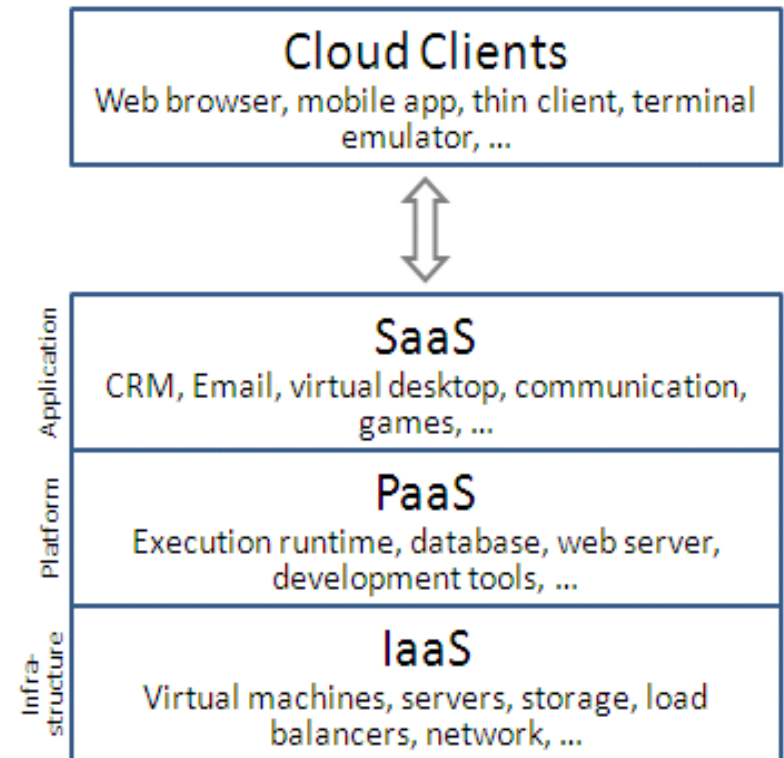
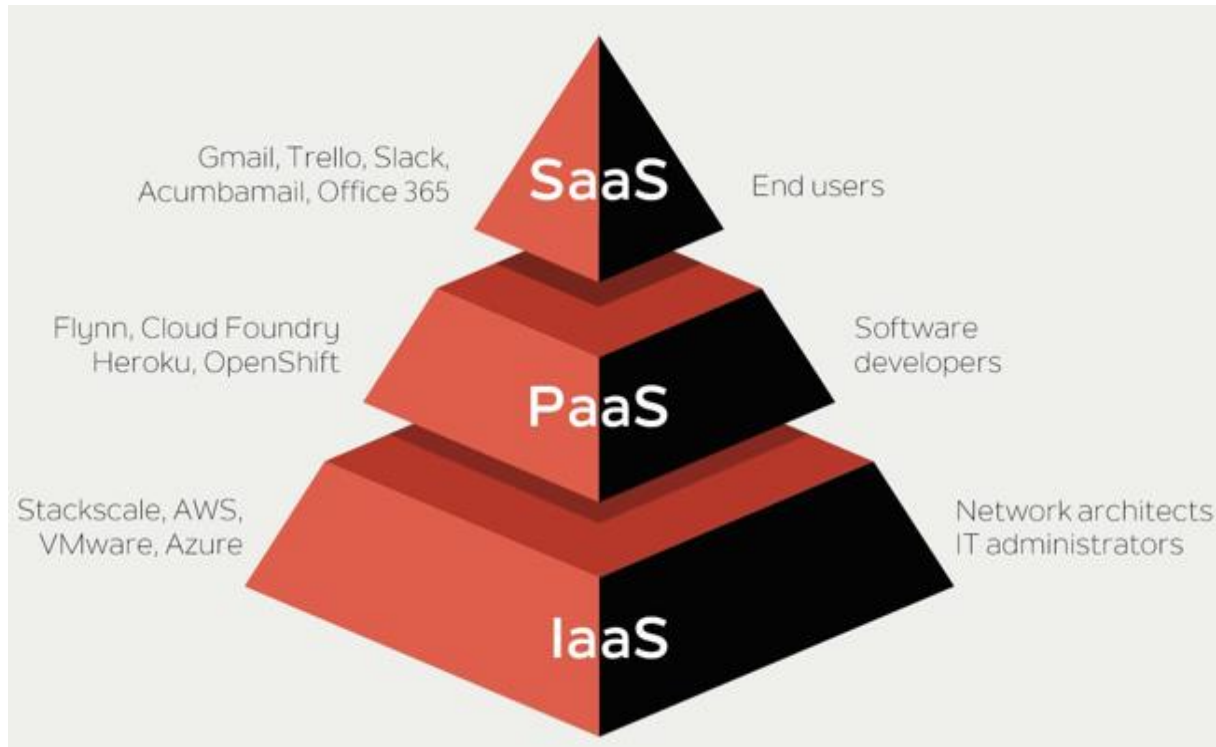
SaaS



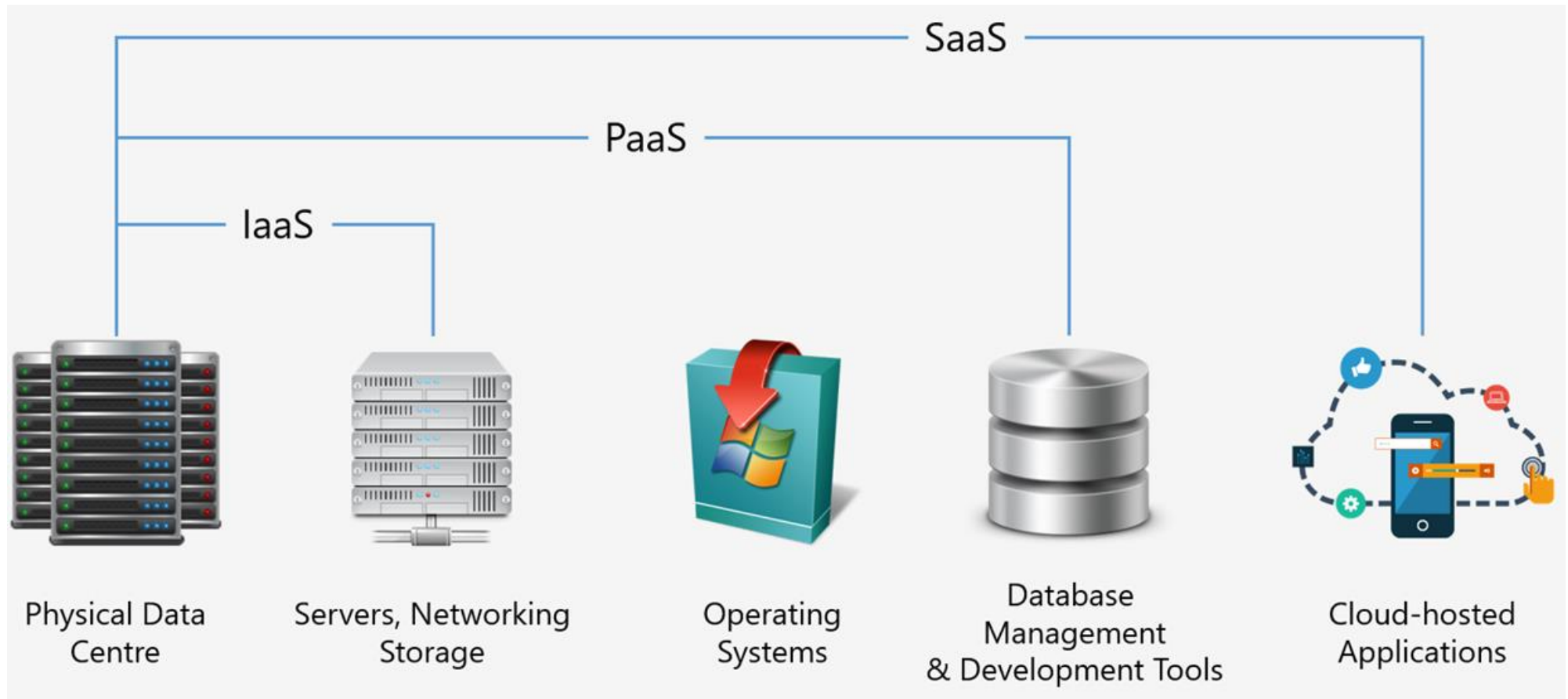
IaaS, PaaS and SaaS

- **On-premises IT infrastructure is like owning a car.** When you buy a car, you're responsible for its maintenance, and upgrading means buying a new car.
- **IaaS is like leasing a car.** When you lease a car, you choose the car you want and drive it wherever you wish, but the car isn't yours. Want an upgrade? Just lease a different car!
- **PaaS is like taking a taxi.** You don't drive a taxi yourself, but simply tell the driver where you need to go and relax in the back seat.
- **SaaS is like going by bus.** Buses have assigned routes, and you share the ride with other passengers.

IaaS, PaaS and SaaS



IaaS, PaaS and SaaS



IaaS, PaaS and SaaS

Basis Of	IAAS	PAAS	SAAS
Stands for	Infrastructure as a service.	Platform as a service.	Software as a service.
Uses	IAAS is used by network architects.	PAAS is used by developers.	SAAS is used by the end user.
Access	IAAS gives access to the resources like virtual machines and virtual storage.	PAAS gives access to run time environment to deployment and development tools for application.	SAAS gives access to the end user.

IaaS, PaaS and SaaS

Basis Of	IAAS	PAAS	SAAS
Model	It is a service model that provides virtualized computing resources over the internet.	It is a cloud computing model that delivers tools that are used for the development of applications.	It is a service model in cloud computing that hosts software to make it available to clients.
Technical understanding.	It requires technical knowledge.	Some knowledge is required for the basic setup.	There is no requirement about technicalities company handles everything.

IaaS, PaaS and SaaS

Basis Of	IAAS	PAAS	SAAS
Popularity	It is popular among developers and researchers.	It is popular among developers who focus on the development of apps and scripts.	It is popular among consumers and companies, such as file sharing, email, and networking.
Percentage rise	It has around a 12% increment.	It has around 32% increment.	It has about a 27 % rise in the cloud computing model.
Usage	Used by the skilled developer to develop unique applications.	Used by mid-level developers to build applications.	Used among the users of entertainment.

IaaS, PaaS and SaaS

Basis Of	IAAS	PAAS	SAAS
Cloud services.	Amazon Web Services, sun, vCloud Express.	Facebook, and Google search engine.	MS Office web, Facebook and Google Apps.
Enterprise services.	AWS virtual private cloud.	Microsoft Azure.	IBM cloud analysis.
Outsourced cloud services.	Salesforce	Force.com, Gigaspaces.	AWS, Terremark

IaaS, PaaS and SaaS

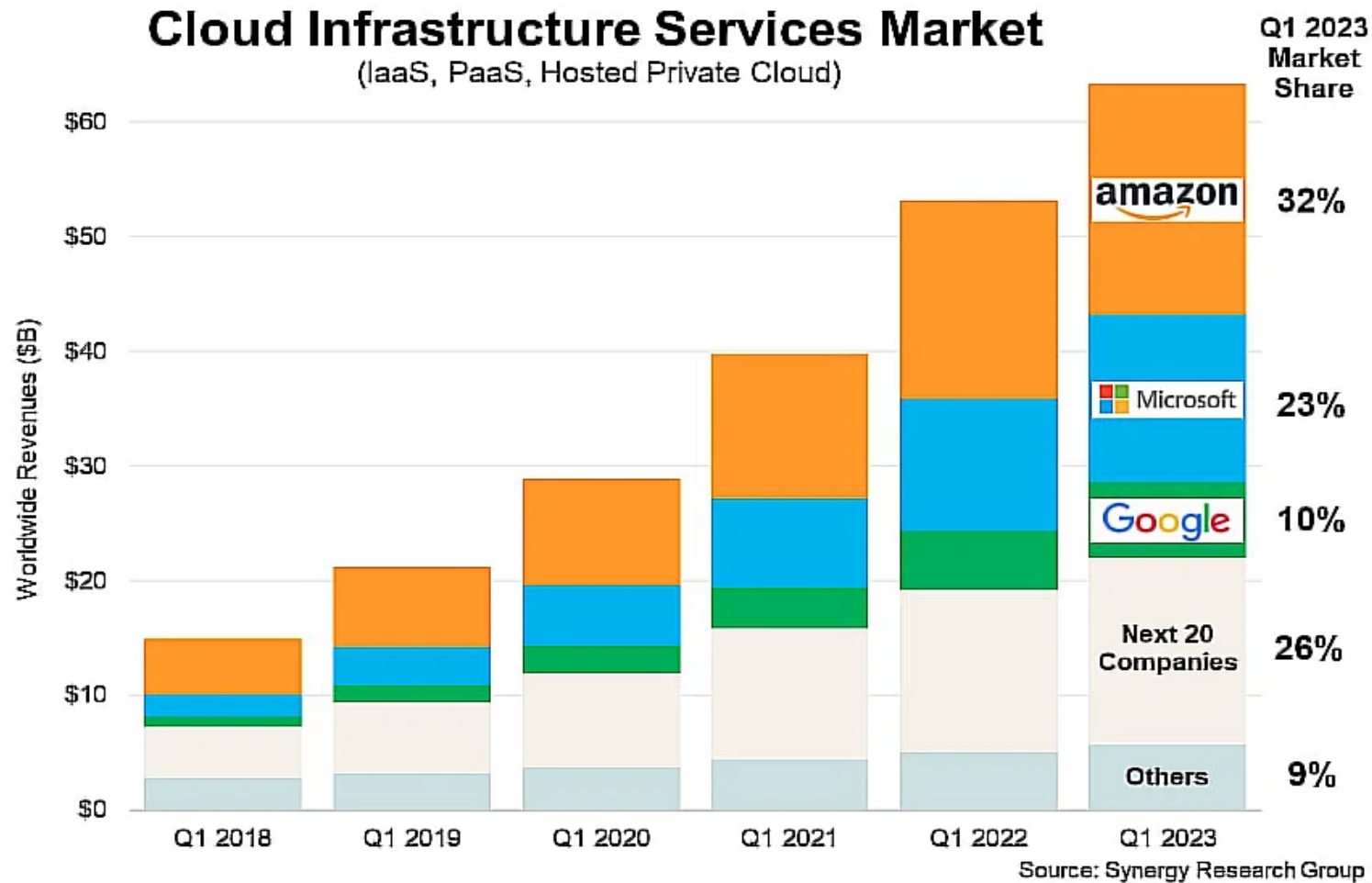
Table 1. Worldwide Public Cloud Services End-User Spending Forecast (Millions of U.S. Dollars)

	2021	2022	2023
Cloud Business Process Services (BPaaS)	51,410	55,598	60,619
Cloud Application Infrastructure Services (PaaS)	86,943	109,623	136,404
Cloud Application Services (SaaS)	152,184	176,622	208,080
Cloud Management and Security Services	26,665	30,471	35,218
Cloud System Infrastructure Services (IaaS)	91,642	119,717	156,276
Desktop as a Service (DaaS)	2,072	2,623	3,244
Total Market	410,915	494,654	599,840

BPaaS = business process as a service; IaaS = infrastructure as a service; PaaS = platform as a service; SaaS = software as a service. Note: Totals may not add up due to rounding.

Source: Gartner (April 2022)

IaaS, PaaS and SaaS



IaaS, PaaS and SaaS

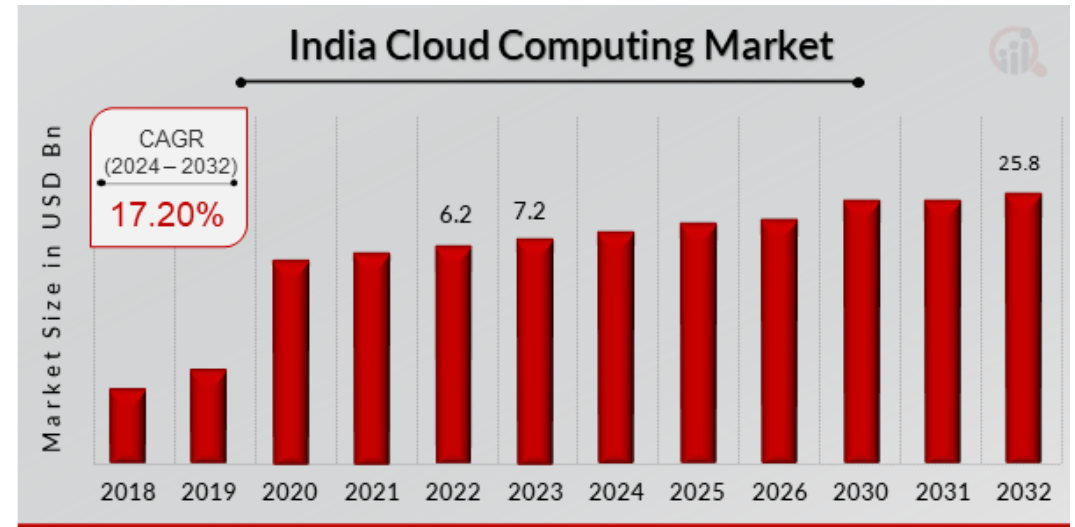
The 50 Biggest SaaS Companies

On U.S. stock exchanges

Last updated: Aug 3, 2023, 7:14 AM

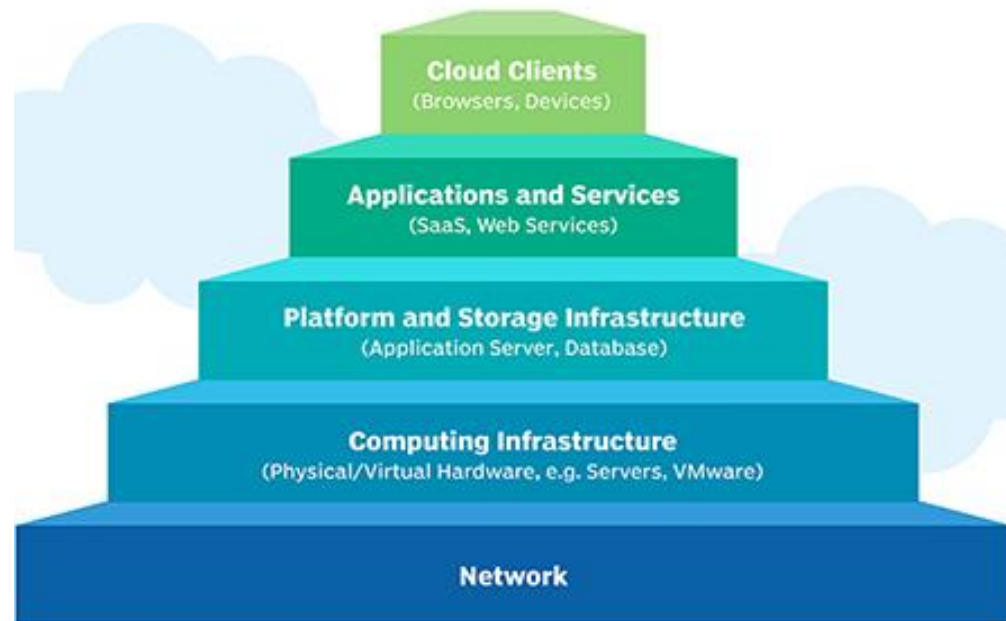
Company (Ticker)	Market Cap
1. Adobe Inc. (ADBE)	\$241.1B
2. Salesforce, Inc. (CRM)	\$219.7B
3. Intuit Inc. (INTU)	\$143.3B
4. ServiceNow, Inc. (NOW)	\$116.0B
5. Shopify Inc. (SHOP)	\$84.5B
6. Workday, Inc. (WDAY)	\$61.3B
7. Snowflake Inc. (SNOW)	\$57.0B
8. Block, Inc. (SQ)	\$47.4B
9. Autodesk, Inc. (ADSK)	\$44.8B
10. Atlassian Corporation (TEAM)	\$44.8B
11. Palantir Technologies Inc. (PLTR)	\$37.7B
12. CrowdStrike Holdings, Inc. (CRWD)	\$36.9B
13. Datadog, Inc. (DDOG)	\$36.1B
14. Veeva Systems Inc. (VEEV)	\$32.3B
15. Samsara Inc. (IOT)	\$29.3B
16. MongoDB, Inc. (MDB)	\$29.1B
17. HubSpot, Inc. (HUBS)	\$27.6B
18. Zscaler, Inc. (ZS)	\$23.0B
19. Cloudflare, Inc. (NET)	\$22.2B
20. Zoom Video Communications, Inc. (ZM)	\$21.7B

1 - 50 / 90 < >

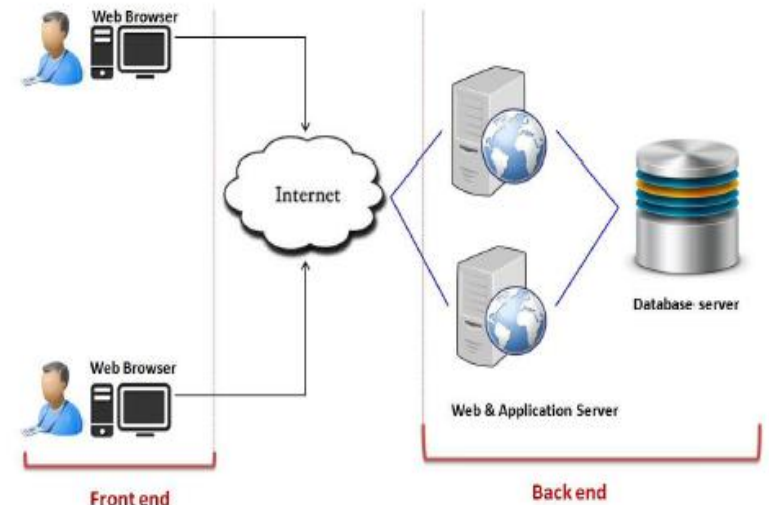
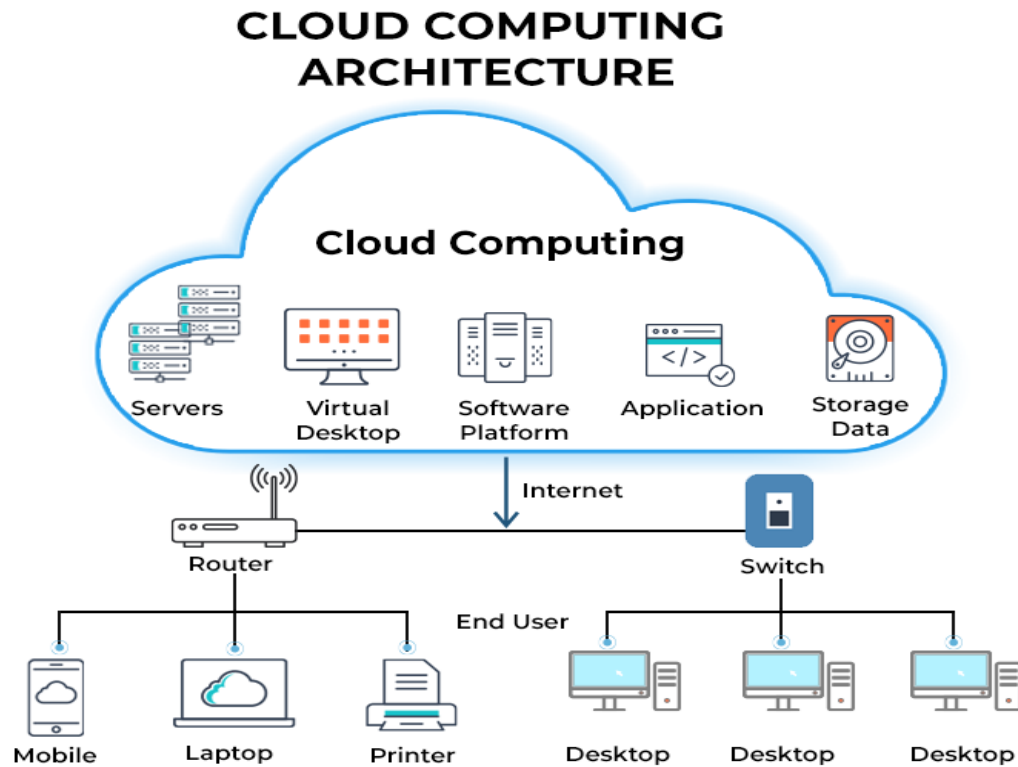


Cloud Computing Architecture

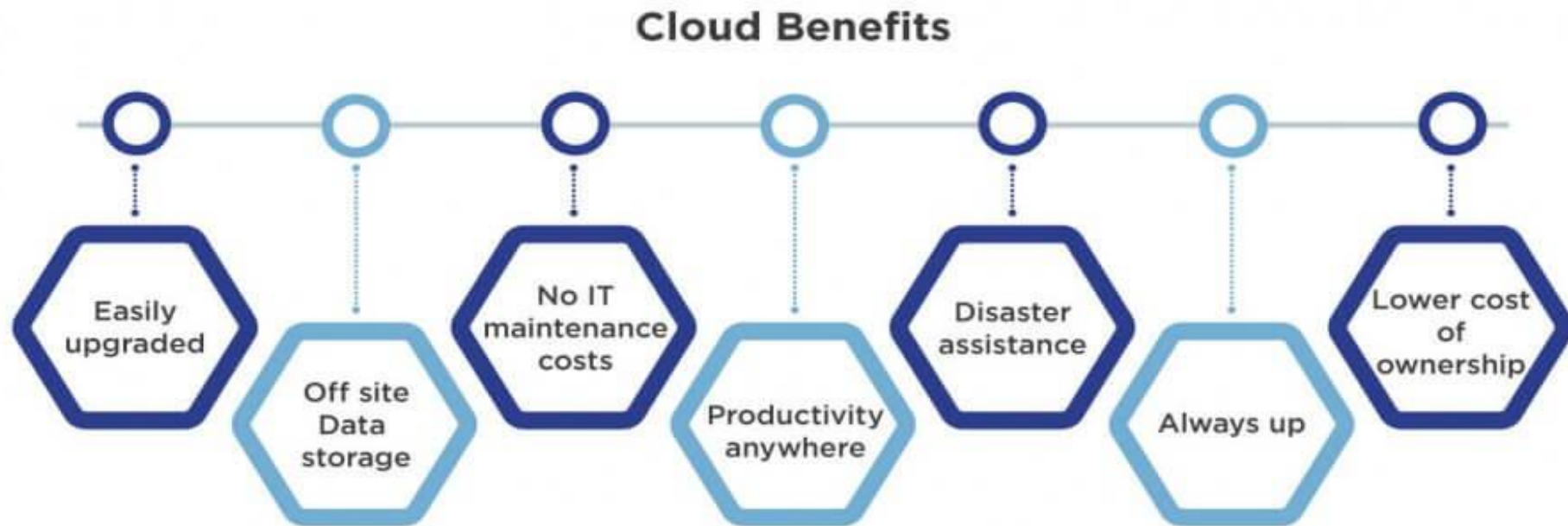
- Cloud infrastructure refers to the hardware and software components, such as servers, storage, a network and virtualization software that are needed to support the computing requirements of a cloud computing model.
- Cloud infrastructure also includes an abstraction layer that virtualizes resources and logically presents them to users through application program interfaces and API-enabled command-line or graphical interfaces.



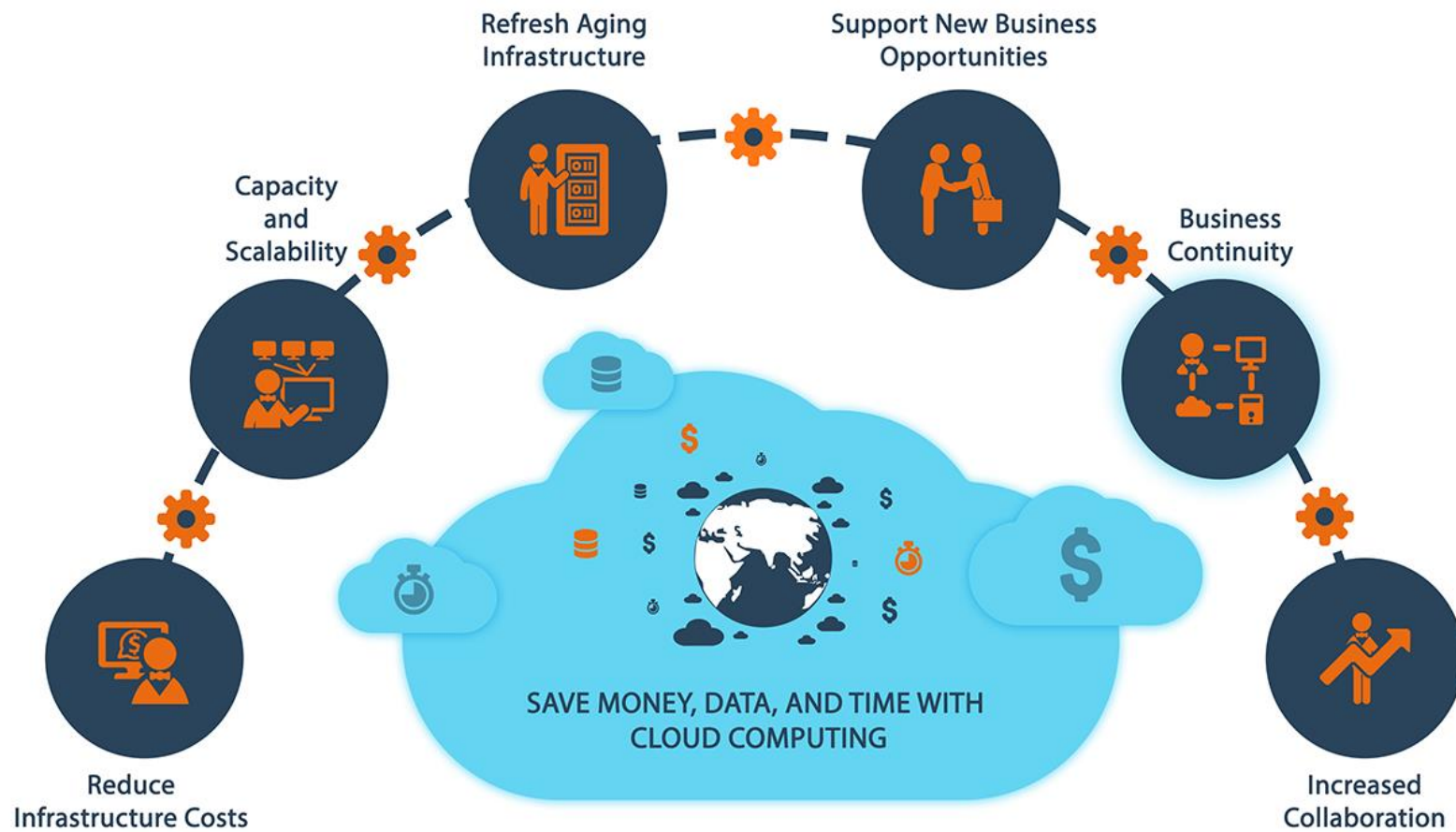
Cloud Computing Architecture



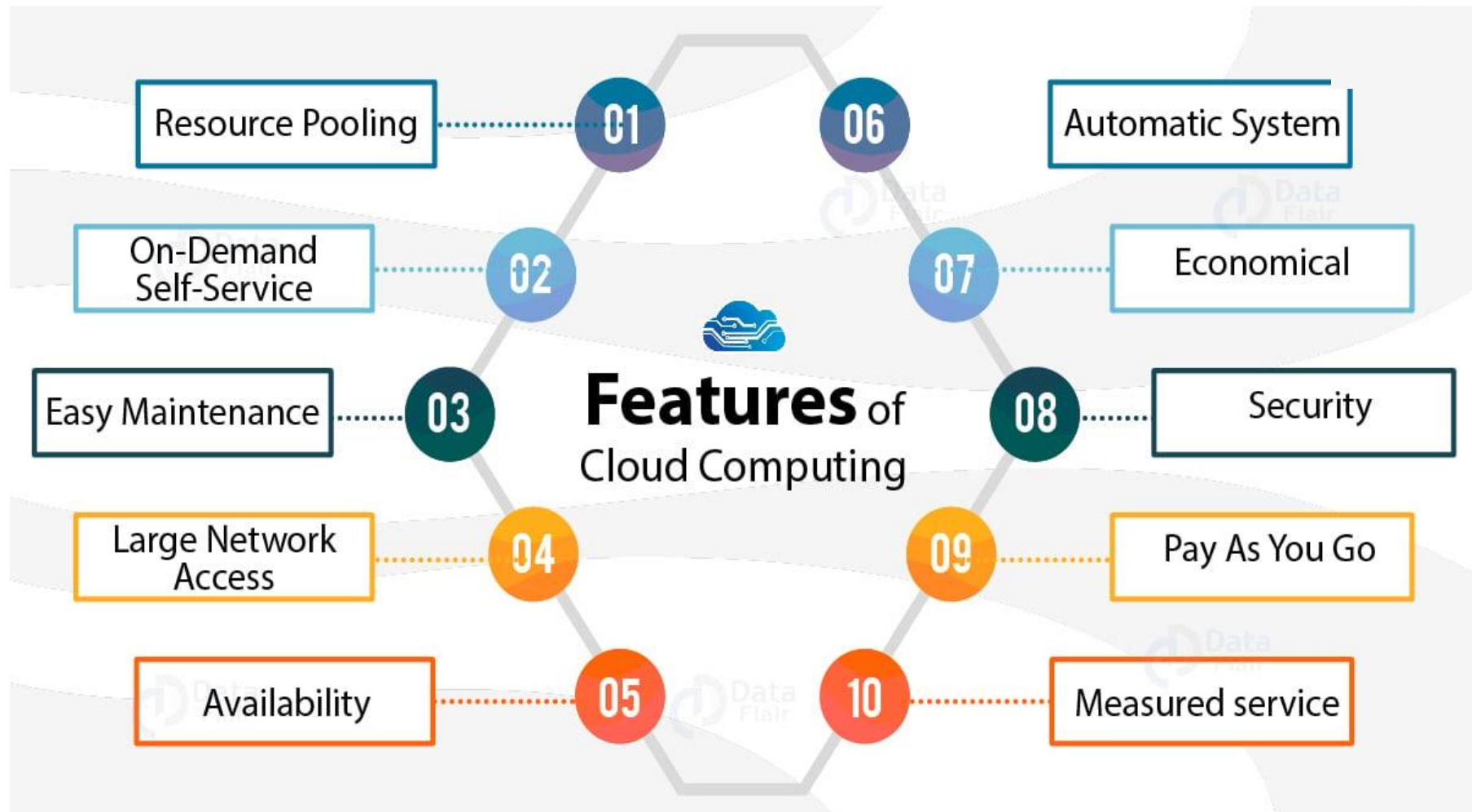
Cloud Computing Benefits



Cloud Computing Benefits



Cloud Computing Features



Cloud Vulnerabilities

- Clouds are affected by malicious attacks and failures of the infrastructure, e.g., power failures.
- Such events can affect the Internet domain name servers and prevent access to a Cloud or can directly affect the Clouds:
 - in 2004 an attack at Akamai caused a domain name outage and a major blackout that affected Google, Yahoo, and other sites.
 - in 2009, Google was the target of a denial of service attack which took down Google News and Gmail for several days;
 - in 2012 lightning caused a prolonged down time at Amazon.

Cloud Threats

- Ease of Use
- Secure Data Transmission
- Insecure APIs
- Malicious Insiders
- Shared Technology Issues
- Data Loss
- Data Breach
- Account/Service Hijacking
- Unknown Risk Profile
- Denial of Service
- Lack of Understanding

Cloud Challenges

1. Availability of service: what happens when the service provider cannot deliver?
2. Data confidentiality and auditability, a serious problem.
3. Diversity of services, data organization, user interfaces available at different service providers limit user mobility; once a customer is hooked to one provider it is hard to move to another.
4. Data transfer bottleneck; many applications are data-intensive.

Cloud Challenges

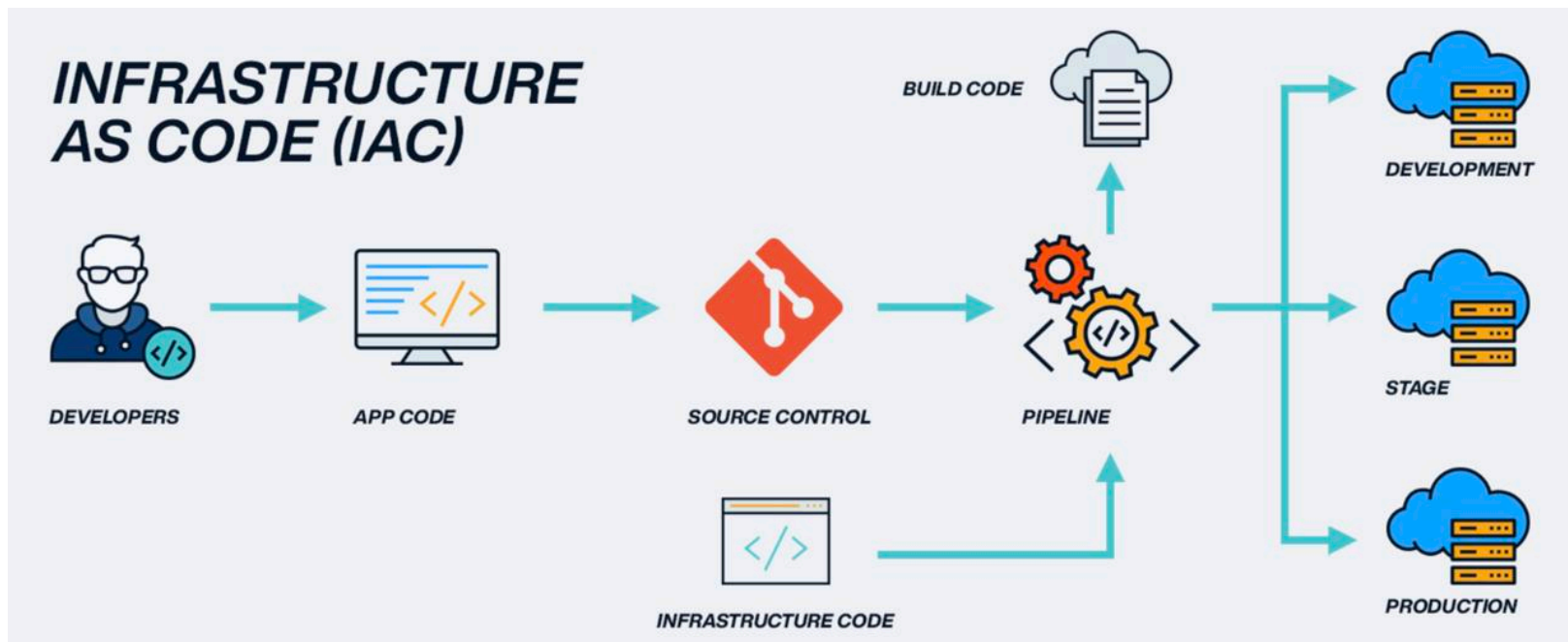
5. Performance unpredictability, one of the consequences of resource sharing.
 - How to use resource virtualization and performance isolation for QoS guarantees?
 - How to support elasticity, the ability to scale up and down quickly?
6. Resource management: It is a big challenge to manage different workloads running on large data centers. Are self-organization and self-management the solution?
7. Security and confidentiality: major concern for sensitive applications, e.g., healthcare applications.

Obstacles of Cloud

- Security concerns
- Data privacy and compliance
- Downtime and service reliability
- Cost management
- Data loss and recovery
- Scalability and performance
- Vendor lock-in
- Lack of cloud expertise

Infrastructure as Code (IaC)

- Infrastructure as code (IaC) is the ability to provision and support user computing infrastructure using code instead of manual processes and settings. Any application environment requires many infrastructure components like operating systems, database connections, and storage.

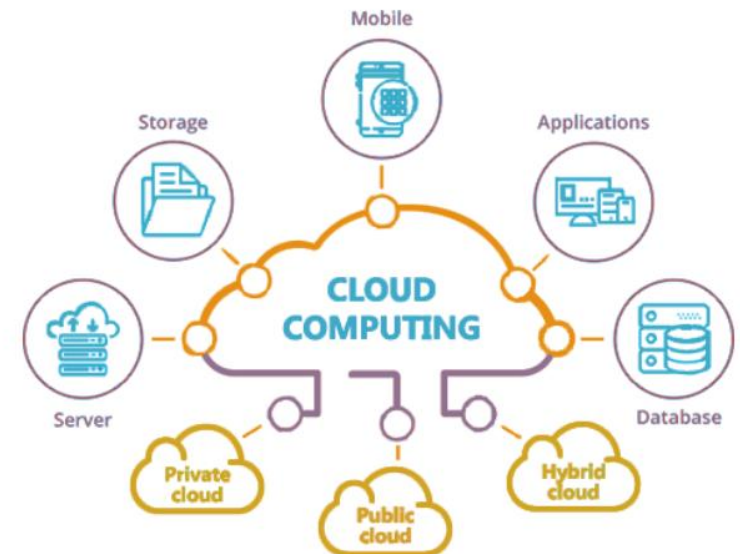


Infrastructure as Code (IaC)

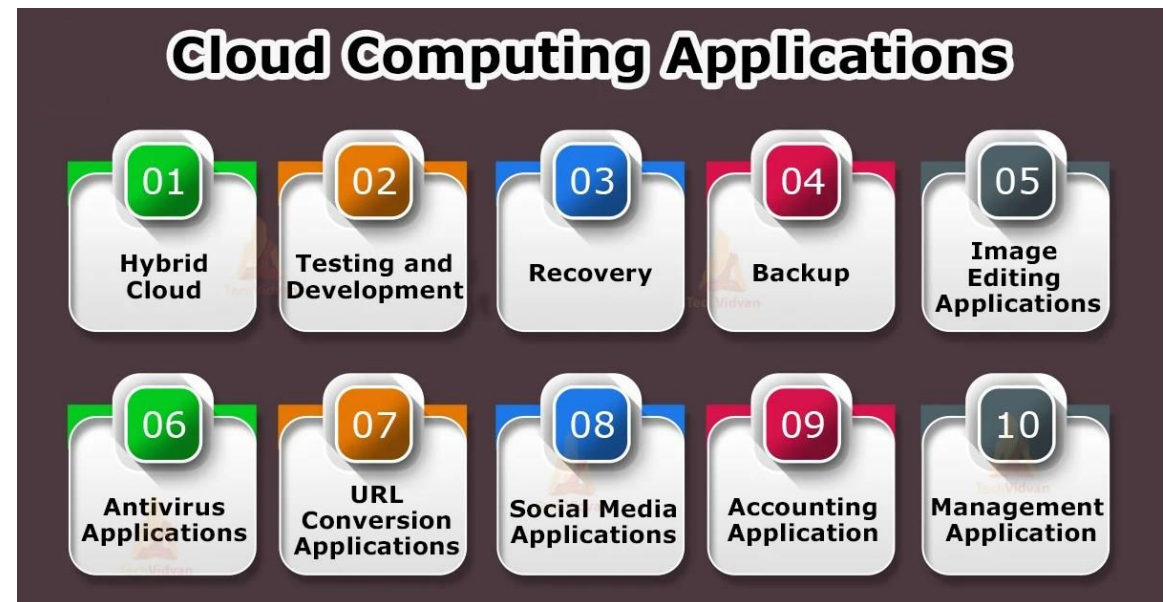
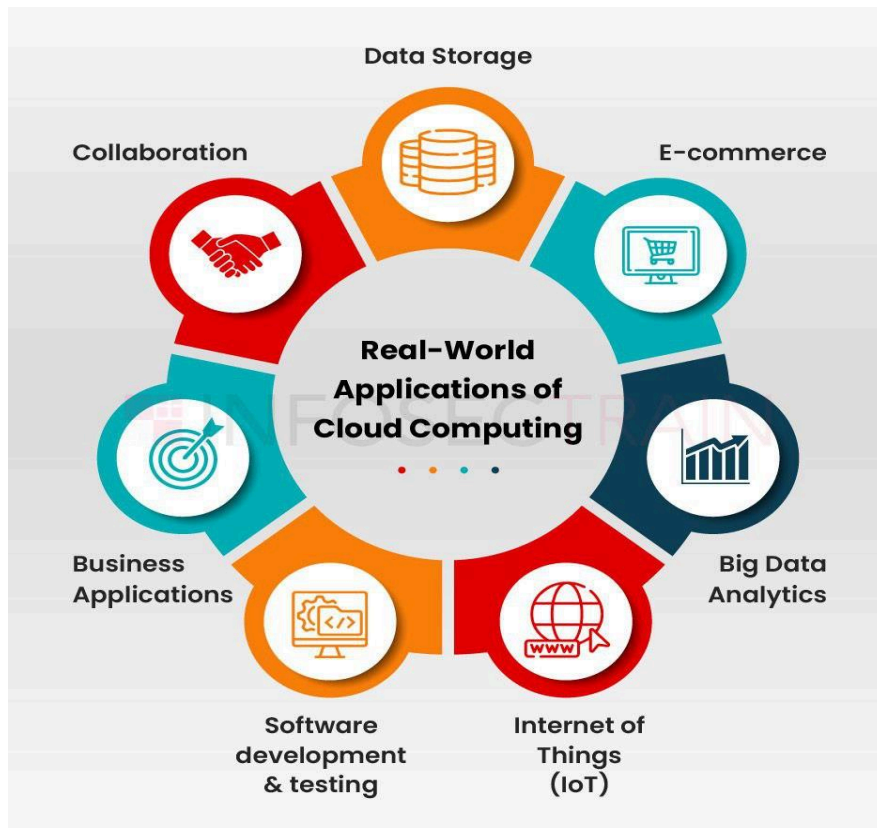
- Infrastructure as code enables organizations to automate tasks and processes that would otherwise be done manually, like managing infrastructure and provisioning resources.
- Treating infrastructure as if it were code lets user adopt powerful practices that have been used by software developers for years, and with great success — practices that include version control, peer review, automated testing, release tagging, release promotion, and continuous delivery.

Practical Applications of Cloud

- Cloud technology has many applications across industries, including business, data storage, management, entertainment, social networking, education, and art. There are three primary types of cloud computing service models: Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS).
- Several cloud platforms are available, each with its own set of features and benefits, such as Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.
- Cloud computing has numerous applications in a variety of industries.



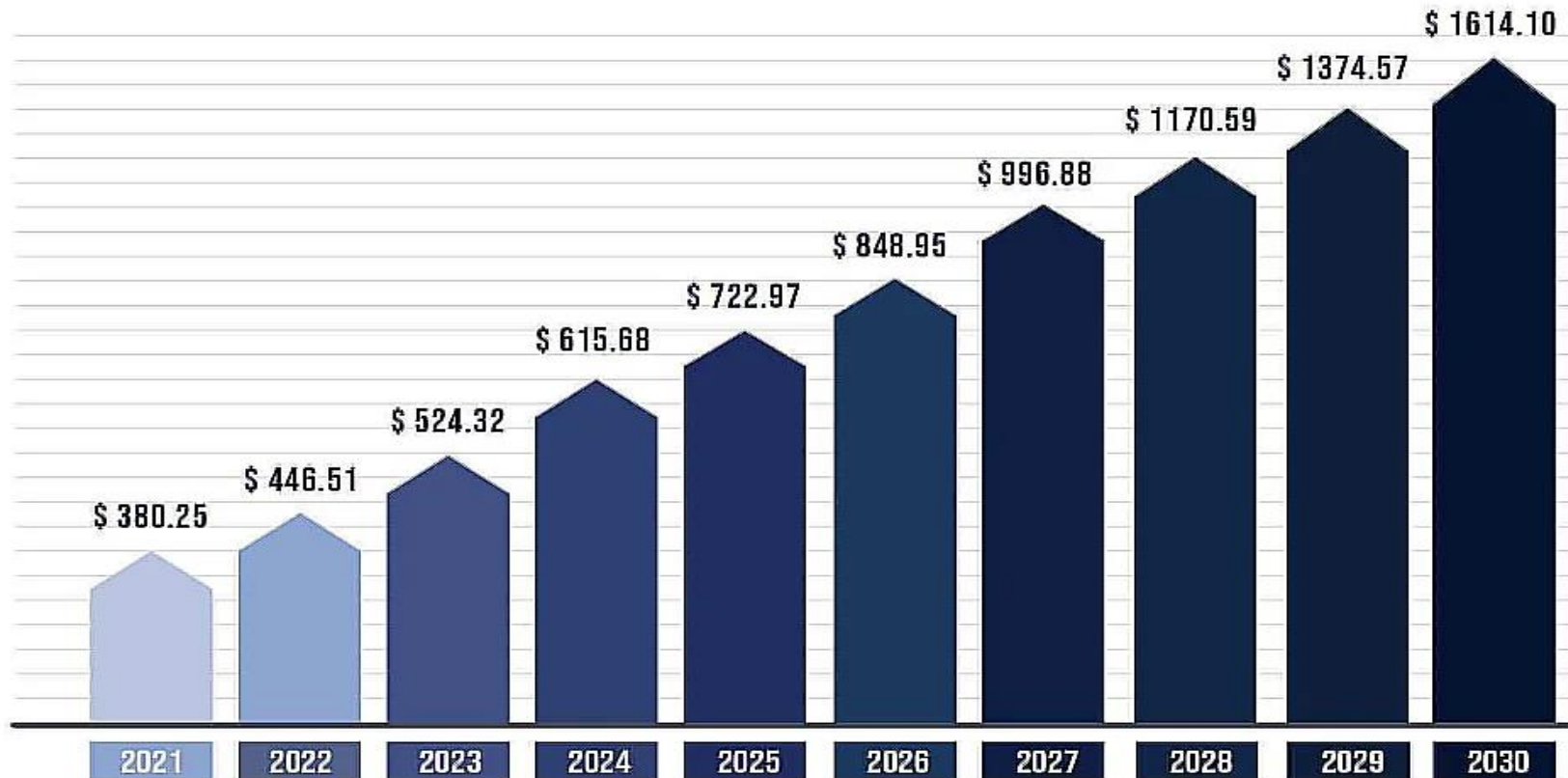
Practical Applications of Cloud



Sample Reports

PRECEDENCE
RESEARCH

CLOUD COMPUTING MARKET SIZE, 2021 TO 2030 (USD BILLION)





References

- Eva Kalyvianaki, Cloud Computing Introduction, University of Cambridge, Lecturing Slides
<http://www.cst.cam.ac.uk/~ek264/>
- <https://www.geeksforgeeks.org/evolution-of-cloud-computing/>
- <https://www.javatpoint.com/history-of-cloud-computing>
- <https://www.seasiainfotech.com/blog/history-and-evolution-cloud-computing/>
- <https://www.javatpoint.com/cloud-service-models>
- <https://www.cloudzero.com/blog/cloud-computing-market-size/>
- <https://www.flackbox.com/cloud-service-models>



JAIN UNIVERSITY

Deemed to be University as per the UGC Act, 1956





THANK YOU